Projection-Pursuit Nearest Neighbors: Distributional Neighborhoods for High-Dimensional Supervised Learning

Jacques Touboul

April 27, 2026

Abstract

The k-Nearest Neighbors algorithm aggregates responses over points close in Euclidean distance, an act that loses statistical power as ambient dimension grows. We propose replacing it by aggregation over the sub-population sharing the same underlying law, identified through a φ-divergence projection-pursuit factorisation (Touboul, 2010, 2025). We define three estimators in this **Projection-Pursuit Nearest Neighbors** (PP-NN) family—Hard, PP-kNN, and Dispatch—and prove three theorems: consistency at parametric rate $n^{-1/2}$ with the effective dimension replacing the ambient one; bounded influence-function sensitivity under Hellinger, which Kullback–Leibler and χ^2 lack; and a Wilks-type χ^2_q null distribution for the factorisation test underwriting the dispatch. A controlled dimension-scaling experiment confirms the central claim: as d grows from 5 to 200 on a Gaussian-mixture design, k-NN's classification error grows fourfold while the distributional-neighborhood estimator stays at $\approx 1\%$, with crossover at $d \approx 30$. The same experiment exposes a prerequisite: the gating step itself suffers a curse of dimensionality, requiring dimension reduction. A real-data benchmark at moderate d shows the PP-NN family does not yet realise this advantage; we identify five theory-practice discrepancies, each paired with a concrete mitigation.

1 Introduction

The metric-to-distributional move

Local non-parametric methods define neighborhoods *geometrically*: the k-Nearest Neighbors (k-NN) algorithm (Cover and Hart, 1967; Fix and Hodges, 1951) predicts at a query point x by aggregating response values over the k training points closest to x in Euclidean distance. This is *a metric act*: it exploits the ambient geometry of $\mathbb{R}^d$, and aggregates responses under the hypothesis that y varies slowly within small Euclidean balls. The hypothesis is not justified by the data-generating process; it is imposed from the outside. In high dimension, it becomes untenable. The well-documented concentration of distances (Beyer et al., 1999) collapses the ratio of nearest to farthest neighbour distances towards 1, so "nearest" loses its discriminative power, and the k-NN estimator converges at the curse-bound rate $O(n^{-2/(d+4)})$, which degrades toward zero as d grows.

We argue that this metric act is conceptually misplaced. When the data admit latent structure—mixture components, regime switches, low-rank organisation—the appropriate act of aggregation is *probabilistic*: pool responses from the sub-population that shares the same underlying law as the query point. Two training points are not "near" because a Euclidean metric said so; they are near because they are i.i.d. draws from the same distributional component. *If* one can identify that component, aggregation becomes statistically grounded: it is the same act as estimating a parametric quantity from a homogeneous sample, and achieves the parametric rate $O(n^{-1/2})$ instead of the metric-based $O(n^{-2/(d+4)})$.

This conceptual move—from metric to distributional neighborhoods—requires a mechanism to identify the components. Mixture Discriminant Analysis (Hastie and Tibshirani, 1996) uses Gaussian likelihood on labelled data; Mixtures of Experts (Jordan and Jacobs, 1994; Yuksel et al., 2012) learn gating by joint likelihood maximisation; Gaussian-mixture + Quadratic-Discriminant-Analysis (GMM+QDA) fits a two-stage Gaussian mixture. These methods share a common fragility: they are *generative* under Gaussian assumptions, and their performance degrades when the data depart from Gaussian mixtures. They also lack a certificate of the mixture hypothesis they rest on.

Projection pursuit, introduced by Friedman and Tukey (1974) and developed by Huber (1985), provides a third route. The φ-divergence formulation of Touboul (2010, 2025) factorises the density f as a known elliptical component g acting on optimal directions $a_1, \ldots, a_k$ plus a residual, with a formal test for the factorisation hypothesis. This framework has three features that address the limitations above: (i) it admits non-Gaussian residual structure via the φ-divergence choice; (ii) it produces a *certificate*—a p-value—of the factorisation adequacy; (iii) it exploits a low-rank representation of the data, in principle escaping the curse of dimensionality.

Contributions

This paper studies the resulting family of local estimators, which we call **Projection-Pursuit Nearest Neighbors** (PP-NN), and asks whether the conceptual argument above translates into practical gains. We contribute:

1. **The PP-NN estimator family.** We define three variants built on Touboul's framework: PP-NN Hard (hard assignment to the dominant component), PP-kNN (hybrid k-NN within the component), and PP-NN Dispatch (a p-value-gated selection between k-NN, PP-NN, and PP-kNN based on the factorisation test). The dispatch rule turns the test statistic into an operational model-selection mechanism—to our knowledge, a new contribution.

2. **Theoretical analysis.** We prove consistency of the PP-NN regressor at the parametric rate $n^{-1/2}$ under the homogeneity assumption, and of PP-kNN at the non-parametric rate $n^{-2/(2+d')}$ where d' is the within-component effective dimension (typically much smaller than d). We derive the influence functions under KL, Hellinger, and χ^2 divergences, showing that Hellinger provides a bounded gross-error sensitivity that the other two lack. We give a Wilks-type proof of the null distribution of the factorisation test (Theorem 3).

3. **Controlled validation of the core claim.** Section 6 reports a dimension-scaling experiment on a $K = 3$ Gaussian-mixture design with d varying from 5 to 200, where the informative signal lives on two directions hidden among $d - 2$ noise directions. The results confirm the theoretical prediction sharply: k-NN's error grows from 0.9% at $d = 5$ to 3.8% at $d = 200$ (a four-fold degradation), while a distributional-neighborhood estimator with dimension reduction stays at $\approx 1\%$ throughout. The crossover occurs at $d \approx 30$.

4. **Refinement: dimension reduction of the gating step is essential.** The same experiment reveals that a naive full-covariance implementation of the distributional estimator fails at high d: the gating step itself suffers from the curse of dimensionality (the mixture parameter count scales as $K \cdot d^2$). The theoretical gain only materialises when the mixture model is fit in a properly identified low-dimensional subspace. This is a methodological prescription: *PP-NN must find the informative directions before it can exploit them.* It is also a new discrepancy we add to the taxonomy.

5. **Real-data benchmark and implementation-limit analysis.** A real-data benchmark (Section 7) on five sklearn datasets at moderate dimensions ($d \leq 30$) positions the PP-NN family against k-NN, Random Forest, GMM+QDA, MDA, and Mixture-of-Experts. At these moderate dimensions the theoretical advantage of PP-NN does not yet materialise: MoE

achieves the best mean rank (2.0), and our Nelder–Mead-based PP optimisation is not sharp enough to realise the population-level gains. We identify five implementation-related theory-practice discrepancies (Section 8) and propose corrections.

How to read this paper

The narrative structure is deliberate. We develop the theoretical case for PP-NN in Sections 2–4, then validate the core theoretical claim empirically under controlled conditions in Section 6, then honestly report the real-data benchmark in Section 7 where implementation limitations become visible, and finally diagnose the gap in Section 8. Readers should interpret the real-data underperformance as *a constraint on the current implementation*, not as a refutation of the distributional-neighborhood principle. The controlled experiment of Section 6 is, in our view, the scientific core of the paper: it shows that the principle is sound, while the real-data benchmark delimits the operational regime in which today's PP-NN implementation realises that soundness.

Paper organization. Section 2 reviews Touboul's φ-divergence PP framework. Section 3 introduces the three PP-NN estimators. Section 4 develops the three central theorems. Section 5 presents two simulation designs that illustrate the method's regimes of success and failure. Section 6 reports the dimension-scaling validation experiment that confirms the central theoretical claim. Section 7 presents the real-data benchmark on five sklearn datasets. Section 8 inventories the five theory-practice discrepancies diagnosed from the real-data benchmark. Section 9 discusses limitations. Section 10 concludes. The companion technical report (Touboul, 2026) contains full proofs, extended simulations, additional figures, and the implementation notes for the `ppnn` Python package.

2 Background: φ-Divergence Projection Pursuit

Let $X \in \mathbb{R}^d$ be a random vector with density f with respect to Lebesgue measure. Following Huber (1985) and Touboul (2010), one seeks a decomposition of the form

$$f(x) = g\big(a_1^\top x, \ldots, a_k^\top x\big) \cdot h(x), \tag{1}$$

where g is a density of prescribed parametric form (e.g., Gaussian or elliptical) on $\mathbb{R}^k$, and h is a residual density satisfying a co-factor condition that leaves g uniquely identified. The directions $a_1, \ldots, a_k \in \mathbb{R}^d$ are unknown and must be estimated from data.

Huber's original proposal maximizes the Kullback–Leibler divergence between f and a Gaussian reference to extract the projections that are *least* Gaussian, leaving a Gaussian core as the residual. Touboul (2010) reformulated the problem in terms of φ-divergences,

$$D_\varphi(P\|Q) = \int \varphi\left(\frac{dP}{dQ}\right) dQ, \tag{2}$$

providing a unified treatment that includes Kullback–Leibler, Hellinger, χ^2, and Cressie–Read divergences as special cases. Estimation proceeds by minimizing the empirical φ-divergence between the data and the hypothesized factorization, yielding estimators $\hat{a}_1, \ldots, \hat{a}_k$ together with a formal test of the null hypothesis that the factorization (1) holds.

The Touboul (2025) extension develops this machinery further in the context of dynamic portfolio construction, where one seeks projections that decorrelate the asset returns in a provable sense. While the financial application is domain-specific, the underlying methodological contribution—identifying distributional sub-structure along learned directions—is general.

Two features of this framework are essential for what follows:

(i) The decomposition is defined on the *entire support* of f: a point x is not assigned hard-wise to g or h, but contributes to both via the factorization. Component membership must therefore be reconstructed a posteriori, typically via responsibilities.

(ii) The framework provides a *testable* claim that a sub-population follows a prescribed law. This distinguishes it from unsupervised clustering (e.g., k-means, Gaussian mixture models in ambient dimension), where component identity is a modeling convenience rather than a statistically certified property.

3 The PP-NN Estimator

3.1 Setup

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled sample with $x_i \in \mathbb{R}^d$ and $y_i \in \mathcal{Y}$ (a finite set for classification, $\mathbb{R}$ for regression). We assume the covariates x_i are drawn i.i.d. from a density f admitting a (possibly iterated) projection pursuit decomposition into K components with parameters $\theta_1, \ldots, \theta_K$, each associated with a direction set $\{a_1^{(\ell)}, \ldots, a_{k_\ell}^{(\ell)}\}$ and a reference law g_ℓ.

3.2 Component responsibilities

For each component $\ell = 1, \ldots, K$, define the posterior responsibility of x:

$$\pi_\ell(x) \;=\; \frac{w_\ell \, g_\ell\Big(a_1^{(\ell)\top} x, \ldots, a_{k_\ell}^{(\ell)\top} x \;;\; \theta_\ell\Big)}{\sum_{m=1}^K w_m \, g_m\Big(a_1^{(m)\top} x, \ldots, a_{k_m}^{(m)\top} x \;;\; \theta_m\Big)}, \tag{3}$$

where $w_\ell \in (0,1)$ is the estimated mixing weight of component ℓ, satisfying $\sum_\ell w_\ell = 1$. This is the standard posterior probability of component membership in a projected mixture model, and it is well-defined because PP delivers the projection directions and component parameters.

3.3 Prediction

Given a query point x, the PP-NN estimator aggregates the responses of the training points weighted by the responsibilities of the component to which each belongs. Two variants are natural.

Soft PP-NN (regression). For continuous y,

$$\hat{y}_{\text{soft}}(x) \;=\; \sum_{\ell=1}^K \pi_\ell(x) \, \bar{y}_\ell, \qquad \bar{y}_\ell \;=\; \frac{\sum_{i=1}^n \pi_\ell(x_i) \, y_i}{\sum_{i=1}^n \pi_\ell(x_i)}. \tag{4}$$

Hard PP-NN. Assign each x_i to $C_{\ell^*(i)}$ where $\ell^*(i) = \arg\max_\ell \pi_\ell(x_i)$, then predict for x by

$$\hat{y}_{\text{hard}}(x) \;=\; \text{aggregate}\big(\{y_i : \ell^*(i) = \ell^*(x)\}\big), \tag{5}$$

using the sample mean for regression or majority vote for classification.

Hybrid PP-kNN. A compromise that recovers locality restricts the metric nearest-neighbor search to the component to which x is primarily assigned:

$$\hat{y}_{\text{hyb}}(x) \;=\; \text{aggregate}\big(\{y_i : x_i \in N_k(x) \cap C_{\ell^*(x)}\}\big). \tag{6}$$

This hybrid variant preserves the in-component smoothness while discarding neighbors that happen to be Euclidean-close but distributionally foreign. We argue in Section 4 that this variant is often the most defensible in practice.

Algorithm 1 PP-NN (hard variant)

Require: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, query point x, reference family g, φ-divergence φ

1: Estimate PP decomposition $\{\hat{a}^{(\ell)}, \hat{\theta}_\ell, \hat{w}_\ell\}_{\ell=1}^K$ by minimizing $\hat{D}_\varphi$ (Touboul, 2010)
2: For each i, compute $\ell^*(i) = \arg\max_\ell \hat{\pi}_\ell(x_i)$
3: For each ℓ, form $C_\ell = \{i : \ell^*(i) = \ell\}$
4: Compute $\ell^*(x) = \arg\max_\ell \hat{\pi}_\ell(x)$
5: **return** aggregate$\big(\{y_i : i \in C_{\ell^*(x)}\}\big)$

4 Statistical Analysis: Three Theorems

This section develops the theoretical core of the paper: three results that justify, respectively, the rate advantage of PP-NN over k-NN in high dimension (Theorem 1), the robustness gain of choosing Hellinger over KL or χ^2 (Theorem 2), and the validity of the p-value-gated dispatch rule (Theorem 3). Each theorem corresponds to one of the three estimators of Section 3: the rate result motivates PP-NN Hard and PP-kNN, the influence-function result motivates the divergence choice in PP-kNN, and the Wilks-type result underwrites PP-NN Dispatch.

The proofs in this section are condensed; full versions, with all technical lemmas and regularity conditions, are in the companion technical report (Touboul, 2026).

4.1 Setup and core assumption

Throughout, $X \in \mathbb{R}^d$ is a covariate vector, $Y \in \mathbb{R}$ (regression) or $Y \in \{1, \ldots, J\}$ (classification) is a response, and we observe $(X_i, Y_i)_{i=1}^n$ i.i.d. from some joint distribution. The PP-NN estimators of Section 3 require the following assumption to deliver their advantages.

Assumption 1 (Componentwise homogeneity)**.** *There exists a latent variable $Z \in \{1, \ldots, K\}$ such that:*

1. *$Y \perp X \mid Z$;*
2. *the conditional distribution of X given $Z = \ell$ admits the PP factorisation (1) with reference law g_ℓ in dimension $k_\ell \ll d$;*
3. *the components are identifiable and well-separated (mixing weights π_ℓ uniformly bounded below, component means non-degenerate).*

Under Assumption 1, the regression function $\eta^*(x) = \mathbb{E}[Y \mid X = x]$ is piecewise constant across components and equal to $\mathbb{E}[Y \mid Z = \ell]$ on component ℓ. The Bayes-optimal predictor is therefore $\eta^*(x) = \sum_\ell \pi_\ell(x)\mathbb{E}[Y \mid Z = \ell]$, the target of the soft PP-NN estimator (4).

4.2 Theorem 1: Rate of convergence in low effective dimension

The first theorem captures the central theoretical claim of the paper: PP-NN converges at a rate that depends on the within-component effective dimension d' rather than on the ambient dimension d.

Theorem 1 (Rate of convergence of PP-NN)**.** *Suppose Assumption 1 holds, the φ-divergence projection-pursuit estimator $\hat{\eta}_n$ converges to η_0 at rate $n^{-1/2}$ (a standard M-estimator result; see van der Vaart, 1998, Ch. 5), and the regression function within each component is β-Hölder smooth. Let $d' := \max_\ell k_\ell$ be the maximal within-component effective dimension. Then:*

1. ***PP-NN Hard.*** *The regression estimator $\hat{\eta}_n^{\text{hard}}(x)$ satisfies, at every continuity point x:*

$$\sqrt{n}\left(\hat{\eta}_n^{\text{hard}}(x) - \eta^*(x)\right) \xrightarrow{d} \mathcal{N}\big(0, \sigma^2_{\text{PP}}(x)\big),$$

where $\sigma^2_{\text{PP}}(x) = \text{Var}(Y \mid Z = \ell^*(x))/\pi_{\ell^*(x)}$ and $\ell^*(x)$ is the dominant component at x.

2. ***PP-kNN.*** *With bandwidth* $k_n \asymp n^{2/(2\beta+d')}$,

$$\mathbb{E}\big[(\hat{\eta}_n^{\mathrm{pp-knn}}(x) - \eta^*(x))^2\big] = O\big(n^{-2\beta/(2\beta+d')}\big).$$

3. ***Comparison with k-NN.*** *The corresponding rate for plain k-NN is* $O(n^{-2\beta/(2\beta+d)})$ *(Stone, 1977). Since* $d' \le d$*, the PP-kNN rate is asymptotically faster, and strictly so when* $d' < d$*.*

Sketch. For (1), apply the delta method to the M-estimator's asymptotic distribution. The PP factorisation reduces the effective parameter dimension from the d-dimensional joint to the d'-dimensional projection plus residual. Under Assumption 1, $Y \perp X \mid Z$ implies $\mathbb{E}[Y \mid X = x]$ is determined entirely by $\pi_\ell(x)$, which depends only on the projected coordinates. Hence the parametric rate transfers from $\hat{\eta}_n \to \eta_0$ to $\hat{\eta}_n^{\mathrm{hard}} \to \eta^*$ via continuous mapping. The asymptotic variance follows from a standard EM-style decomposition.

For (2), the within-component k-NN regressor is consistent at the standard non-parametric rate (Stone, 1977; Devroye et al., 1994) but evaluated in the d'-dimensional projected coordinate system rather than the full ambient space. The composition of consistent component-assignment ($\hat{\pi}_\ell \to \pi_\ell$ uniformly, by uniform consistency of M-estimators) with consistent within-component k-NN yields the stated rate.

For (3), this is a direct comparison: $-2\beta/(2\beta+d') > -2\beta/(2\beta+d)$ whenever $d' < d$. □ □

Numerical illustration. For $\beta = 1$ (Lipschitz response), $d = 200$, $d' = 2$, $n = 1000$: k-NN rate is $n^{-2/202} \approx 0.93$, PP-kNN rate is $n^{-2/4} \approx 0.032$. The theoretical ratio is ≈ 30. The empirical experiment of Section 6 measures a ratio of ≈ 4. The gap between theoretical and empirical ratios is explained by the constants in the rates—ignored in asymptotic analysis but binding at $n = 1000$—and by the fact that GMMkNN-PCA is an oracle-upper-bound substitute that exposes the population-level rate but not the implementation overhead. The *direction* of the predicted advantage is confirmed.

4.3 Theorem 2: Bounded influence function under Hellinger

The second theorem characterises robustness. It shows that the gross-error sensitivity (the supremum of the influence function, see Hampel, 1974) of the PP-NN estimator is bounded under the Hellinger divergence, but unbounded under KL or χ^2.

For a parametric estimator $\hat{\theta}$ at the true law P_0, the influence function $\mathrm{IF}(x; \hat{\theta}, P_0)$ measures the Gateaux derivative of $\hat{\theta}$ along a contamination $\delta_x - P_0$, and the gross-error sensitivity is $\gamma^* = \sup_x \|\mathrm{IF}(x; \hat{\theta}, P_0)\|$.

Theorem 2 (Influence-function ordering of PP-NN)**.** *Let* $\hat{\eta}_\varphi$ *be the* φ*-divergence projection-pursuit estimator under one of the three divergences:*

- *Kullback–Leibler:* $\varphi_{\mathrm{KL}}(t) = t \log t - t + 1$*;*
- *Hellinger:* $\varphi_{\mathrm{H}}(t) = (\sqrt{t} - 1)^2$*;*
- *Pearson* χ^2*:* $\varphi_{\chi^2}(t) = (t-1)^2/2$*.*

Let g_{η_0} *be a Gaussian reference with finite second moments. The gross-error sensitivities of* $\hat{\eta}_\varphi$ *at* g_{η_0} *satisfy:*

1. $\gamma^*_{\mathrm{H}} < \infty$*;*
2. $\gamma^*_{\mathrm{KL}} = +\infty$ *if the score* $\nabla_\eta \log g_{\eta_0}$ *is unbounded (true for Gaussian);*
3. $\gamma^*_{\chi^2} = +\infty$ *if* $\sup_x g_{\eta_0}(x)^{-1}$ *is unbounded (also true for Gaussian).*

The corresponding sensitivities of the downstream PP-NN estimator inherit the boundedness in (1) and the unboundedness in (2)–(3) (Lipschitz extension via the responsibility map).

Sketch. Following Basu et al. (2011, Sec. 4.4), the influence function of an M-estimator solving $\mathbb{E}_P[\psi(X;\theta)] = 0$ at $P_0 = g_{\eta_0}$ is

$$\mathrm{IF}(x; \hat{\theta}_\varphi, P_0) = -[J(\theta_0)]^{-1}\psi_\varphi(x;\theta_0),$$

where $J(\theta) = \nabla_\theta \mathbb{E}[\psi]$ is the asymptotic Jacobian and ψ_φ is the divergence-specific score function. For Hellinger, $\psi_{\mathrm{H}}(x;\theta) = \nabla_\theta \log g_\theta(x) \cdot (\sqrt{f(x)/g_\theta(x)} - 1)$, which is bounded in x because $\sqrt{f/g}$ is bounded by the Cauchy–Schwarz inequality whenever $\int f\,dx = 1$. For KL, $\psi_{\mathrm{KL}}(x;\theta) = \nabla_\theta \log g_\theta(x)$ is the score, unbounded for the Gaussian. For χ^2, $\psi_{\chi^2}(x;\theta) = \nabla_\theta \log g_\theta(x) \cdot f(x)/g_\theta(x)^2$, unbounded for Gaussian g with light tails.

The PP-NN estimator at a query x_0 is the composition of $\hat{\eta}_\varphi$ with the responsibility map $x \mapsto \pi_\ell(x;\hat{\eta}_\varphi)$, which is Lipschitz under (R2)–(R5). Composition of a Lipschitz function with a bounded-IF estimator yields a bounded-IF estimator. The same composition with an unbounded-IF estimator yields unbounded IF. □ □

What the theorem predicts and what is observed. Theorem 2 predicts that contamination affects KL- and χ^2-based PP-NN unboundedly while Hellinger-based PP-NN is robust. Section 7.2 reports the empirical contamination experiment in which the three divergences produce indistinguishable curves. The diagnostic in Section 4.5 (Discrepancy 3) explains why: the theorem describes the influence function *at* the true minimiser η^*_φ, and our optimiser does not reach it.

4.4 Theorem 3: Null distribution of the factorisation test

The third theorem justifies PP-NN Dispatch. The factorisation test of Touboul (2010, 2025) produces a p-value for the null hypothesis "f admits the PP factorisation $f = g_{\eta_0} \cdot h$". We need to know the null distribution of the test statistic to use the p-value as a gatekeeper.

Theorem 3 (Wilks-type null distribution)**.** *Let $\hat{T}_n = \frac{2n}{\varphi''(1)} \cdot D_\varphi(\hat{f}_n \,\|\, g_{\hat{\eta}_n})$, with $\hat{f}_n$ a kernel density estimator with bandwidth $h_n \to 0$ and $nh_n^{d^*} \to \infty$ (d^* the projected effective dimension), and $\hat{\eta}_n$ the M-estimator of the parameter under H_0. Suppose $\varphi \in \mathcal{C}^3$ with $\varphi(1) = \varphi'(1) = 0$. Then under $H_0 : f = g_{\eta_0}$, $\eta_0 \in \mathcal{E}$:*

$$\hat{T}_n \xrightarrow{d} \chi^2_q \quad \text{as } n \to \infty,$$

where $q = K(d^* + 2) - 1$ is the effective dimension of the parametric family at η_0. Moreover, under any fixed alternative $f_1 \notin \mathcal{F}_K$, $\hat{T}_n \to \infty$ in probability, so the test is consistent.

Sketch. The argument proceeds in four steps; full details are in (Touboul, 2026, App. A).

Step 1 (Taylor expansion). For $\varphi \in \mathcal{C}^3$ with $\varphi(1) = \varphi'(1) = 0$, $D_\varphi(f \,\|\, g_\eta) = \frac{\varphi''(1)}{2} \int (f - g_\eta)^2/g_\eta\,dx + O(\|f - g_\eta\|_\infty^3)$. Under bandwidth conditions, the cubic remainder is $o_{\mathbb{P}}(n^{-1})$.

Step 2 (Quadratic decomposition). Substituting and rescaling, $\hat{T}_n = n \int (\hat{f}_n - g_{\hat{\eta}_n})^2/g_{\hat{\eta}_n}\,dx + o_{\mathbb{P}}(1)$. Decomposing $\hat{f}_n - g_{\hat{\eta}_n} = (\hat{f}_n - g_{\eta_0}) - (g_{\hat{\eta}_n} - g_{\eta_0})$ and Taylor-expanding $g_{\hat{\eta}_n}$ in η, the leading term becomes a squared norm of $Z_n := \sqrt{n}(\hat{f}_n - g_{\eta_0})$ projected orthogonally to the score subspace.

Step 3 (Rank of the score). The Fisher information at η_0 has rank $q = K(d^* + 2) - 1$ for the Gaussian PP factorisation, by parameter counting: K component means in d^* dimensions, K projection-direction parameters on S^{d^*-1}, $K - 1$ free mixing weights.

Step 4 (Weak convergence). The empirical-process limit of Z_n is a centered Gaussian element of $L^2(g_{\eta_0})$ with diagonal covariance kernel. By the continuous mapping theorem, the squared

norm of its projection onto the orthogonal complement of the q-dimensional score subspace converges in distribution to χ^2_q.

For consistency under fixed alternatives, $D_\varphi(f_1 \,\|\, g_\eta) > 0$ for all $\eta \in \mathcal{E}$ by identifiability, and $\inf_\eta D_\varphi(f_1 \,\|\, g_\eta) > 0$ by continuity and compactness, so $\hat{T}_n \to \infty$. □ □

Operational consequence. Theorem 3 provides a calibrated gatekeeper: reject H_0 at level α if $\hat{T}_n > \chi^2_{q,1-\alpha}$. PP-NN Dispatch uses this rule with $\alpha = 0.05$. The test is asymptotically valid, but Section 4.5 (Discrepancy 2) discusses why finite-sample miscalibration and the gap between "factorisation accepted" and "PP-NN beats k-NN" make this gatekeeper imperfect on real data.

4.5 Differences between theory and empirical results: a structural diagnostic

The three theorems above are mathematically correct under their stated assumptions, yet our empirical results (Sections 7 and 8) reveal that they do not deliver the predicted advantages on real data at the moderate sample sizes and dimensions of our benchmark. We pause before the empirics to explain *why*, mechanically. Each of the five theory-practice discrepancies catalogued in Section 8 has the same underlying structure: *a theorem that requires a convergence which the implementation does not realise at finite n.* We make this explicit here so that the empirical sections can be read with the right interpretation.

Discrepancy 3: optimiser insufficiency (theorem → practice gap on Theorem 2). The influence-function ordering Hellinger $\prec$ KL $\prec \chi^2$ is a *population statement*: it concerns the asymptotic Gateaux derivative of $\hat{\eta}_\varphi$ *at the true minimiser* η^*_φ. Mathematically, for a M-estimator,

$$\mathrm{IF}(x; \hat{\theta}_\varphi, P_0) = -[J(\theta^*_\varphi)]^{-1}\, \psi_\varphi(x; \theta^*_\varphi),$$

which is well-defined only when the empirical minimiser $\hat{\theta}_n$ has reached θ^*_φ. Our implementation uses Nelder–Mead with three random restarts on S^{d-1} and 200 iterations each, returning a local minimiser $\hat{\theta}^{\mathrm{local}}_n$ rather than θ^*_φ. The resulting decomposition

$$\hat{\eta}^{\mathrm{KL}}_n - \hat{\eta}^{\mathrm{H}}_n = \underbrace{(\hat{\eta}^{\mathrm{KL}}_n - \eta^*_{\mathrm{KL}})}_{O(\varepsilon_{\mathrm{opt}})} + \underbrace{(\eta^*_{\mathrm{KL}} - \eta^*_{\mathrm{H}})}_{O(\delta_\varphi)} + \underbrace{(\eta^*_{\mathrm{H}} - \hat{\eta}^{\mathrm{H}}_n)}_{O(\varepsilon_{\mathrm{opt}})}$$

shows that the divergence-specific signal δ_φ is observable only when $\varepsilon_{\mathrm{opt}} \ll \delta_\varphi$. In our experiments $\varepsilon_{\mathrm{opt}} \approx 10^{-3}$ (Nelder–Mead tolerance) while $\delta_\varphi \approx 10^{-4}$ (typical inter-divergence gap on Gaussian-mixture data); the optimiser noise dominates. The Riemannian Newton method advocated by Touboul (2025), which exploits the curvature of S^{d-1} and converges quadratically, would reduce $\varepsilon_{\mathrm{opt}}$ to $O(10^{-8})$ and reveal the predicted ordering.

Discrepancy 2: dispatch miscalibration (theorem → practice gap on Theorem 3). The χ^2_q null distribution is asymptotic, with rate of convergence depending on n, d, K, and the bandwidth schedule of the kernel-density estimator. At $n = 178$ (wine dataset), $K = 2$, ambient $d = 13$, the finite-sample distribution of $\hat{T}_n$ can deviate substantially from χ^2_q. Our parametric bootstrap with $B = 200$ resamples then estimates a critical value with itself non-trivial variance (≈ 0.015 at the 5% level). The combined effect is a test whose effective Type-I error departs from the nominal 5% in a direction that depends on the dataset.

A second, more conceptual mismatch compounds the calibration issue. The test answers "does f_X admit the PP factorisation?"—a question about the marginal distribution of X alone. The dispatch rule treats acceptance of H_0 as evidence that "PP-NN will outperform k-NN at predicting Y". These two events are not the same, and on wine in particular they are anti-correlated: the test accepts H_0 on 2 folds of 5, but on those folds PP-NN's classification error is ≈ 0.39, vastly worse than k-NN's ≈ 0.03. The factorisation test cannot see this because it has no

access to Y (Discrepancy 4 below). *Theorem 3 is correct about what it tests; it is the operational interpretation of acceptance that is unjustified.*

Discrepancy 1: structural K-insufficiency (assumption gap on all three theorems). Theorems 1–3 all assume that K is correctly specified (or at least bounded above by the true number of classes). When $K = 2$ and the true label space has $J = 3$ classes (e.g., iris, wine), Assumption 1 fails by construction: the components cannot be aligned with the class structure, so $Y \perp X \mid Z$ does not hold. The piecewise-constant Bayes optimal $\eta^*(x)$ requires at least J pieces; with $K < J$, the best PP-NN-Hard predictor mixes class labels within each component, achieving error close to the majority-class baseline rather than zero. *This is a representational failure, not a statistical one*: more data cannot fix it, only setting $K \geq J$ can.

Discrepancy 4: unsupervised gating (architectural gap on the framework). The PP factorisation optimises a divergence between f_X and $g_\eta \cdot h$, which is a marginal-on-X criterion. The gating directions $a_1, \ldots, a_k$ are chosen to make f_X factorise nicely, not to predict Y well. When $\mathbb{E}[Y \mid X = x]$ depends on a feature direction *orthogonal* to the dominant PP projection direction, PP-NN's components are uninformative for Y. This is the architectural cousin of Discrepancy 2 (the test ignores Y) and the most fundamental obstacle to PP-NN's competitiveness: any successor variant in the family must include Y in its gating objective. Mixture-of-Experts does this by construction (joint likelihood maximisation on (X, Y)), which is one of two reasons it dominates our real-data benchmark.

Discrepancy 5: gating curse of dimensionality (theorem → practice gap on Theorem 1). Theorem 1 states that the relevant dimension is the within-component effective d', not the ambient d. This is a guarantee about the *rate*; it requires that $\hat{\eta}_n$ achieves $n^{-1/2}$ convergence, which itself requires the M-estimator's Fisher information to be invertible. Fitting a K-component Gaussian mixture with full $d \times d$ covariance matrices in ambient space requires estimating $K \cdot d(d+1)/2$ free parameters; at $K = 3$, $d = 80$ that is $\approx 9{,}700$ parameters from 1,000 training points, and the empirical Fisher information becomes ill-conditioned. The mixture fit then converges to a near-singular local minimum and PP-NN inherits the corruption. The remedy—identifying the informative subspace before fitting the mixture—is precisely what Touboul's projection pursuit was designed to do, but it requires running the PP step properly (Discrepancy 3) rather than fitting a mixture in ambient space.

The structural pattern. All five theory-practice discrepancies share a common shape: an asymptotic theorem holds at the population level (Theorems 1–3), and the gap to practice originates in the implementation's inability to realise the convergence the theorem assumes. The optimiser does not reach the divergence-specific minimum (Discrepancy 3); the test statistic does not reach its χ^2 limit at finite n (Discrepancy 2); the model-selection assumption $K \geq J$ is violated (Discrepancy 1); the unsupervised criterion does not align with the supervised target (Discrepancy 4); the in-ambient-space mixture fit is not identifiable (Discrepancy 5). Each discrepancy admits a concrete fix, listed in Table 4, none of which alters the underlying theorems but each of which is needed for the theorems' guarantees to translate to real data. The empirical sections of the paper are best read in this light: as documenting where the implementation currently sits relative to the theorems' prerequisites, not as evidence for or against the theorems themselves.

4.6 Relation to existing methods

PP-NN sits at the intersection of three streams. *Local metric methods* (Cover and Hart, 1967; Fix and Hodges, 1951; Weinberger and Saul, 2009) define neighbourhoods geometrically; PP-NN replaces the metric by a distributional criterion. *Generative mixture models* (Hastie and

Tibshirani, 1996; Scrucca et al., 2016; with the closest classical ancestor Calò, 2007, who fits a GMM after a non-Gaussianity-based PP) use Gaussian likelihoods; PP-NN uses φ-divergences and admits a formal factorisation test. *Conditional mixtures and mixtures of experts* (Jacobs et al., 1991; Jordan and Jacobs, 1994; Yuksel et al., 2012; Masoudnia and Ebrahimpour, 2014) jointly train gating and experts on (X, Y); PP-NN's gating is unsupervised on X alone, which is a deliberate design choice (it inherits the factorisation test from Touboul, 2025) and an empirical liability (Discrepancy 4 in Section 8). The closest distributional-input nearest-neighbours method is Sabri et al. (2024)'s DC-KNN, which takes entire distributions as inputs; PP-NN derives the distributional component from ordinary vector inputs.

5 Simulation Study

Before tackling real data, we verify on two designs that the three estimators of Section 3 behave as the theory of Section 4 predicts. Both designs are 2-dimensional so they can be visualised, but they exhibit fundamentally different alignment between the latent mixture structure and the response.

Setup. For PP-NN and PP-kNN we set $K = 2$ components, optimise the projection direction by maximising the empirical φ-divergence over 40 random initialisations on the unit sphere, then fit a two-component Gaussian mixture on the projected sample. The neighbourhood size is $k = 15$ throughout. We split the sample 70/30 for train and test.

Synthetic design (Assumption 1 holds). We generate $n = 1500$ observations in $\mathbb{R}^2$ from a two-regime mixture: regime 0 is a Gaussian convolved with a Gumbel shift, regime 1 is a separated Gaussian. The response is regime-specific (mean 2 on regime 0, mean -1 on regime 1, with Gaussian noise). PP recovers the latent regimes exactly. The bootstrap factorisation test of Theorem 3 accepts $K = 2$ ($\hat{p}_n = 0.38$ under KL, 0.60 under Hellinger) and rejects $K = 1$ ($\hat{p}_n < 0.005$), credentialing PP-NN. *Result:* on regression, PP-NN achieves test MSE 0.244 versus k-NN's 0.264 (-7.6%); on classification, PP-NN achieves error 0.012 versus k-NN's 0.027 (-56%). The three divergences yield indistinguishable results, consistent with the fact that they share the same minimiser at the true factorisation point.

Financial design (Assumption 1 fails). We then simulate a regime-switching daily return series with $n = 2000$ observations, $d = 4$ covariates (lagged returns and volatility), and a stress regime drawn from a Student-t_4 rather than a Gaussian. The response is the next-day return (regression) or its sign (classification). The factorisation test *rejects* the Gaussian-mixture assumption at $K = 2$ for all three divergences ($\hat{p}_n < 0.005$), correctly signalling that the data lie outside $\mathcal{F}_2$. *Result:* on regression, the three estimators are within 1% of each other (the homogeneity assumption is at most weakly violated for the conditional mean); on classification, PP-NN Hard underperforms k-NN by ≈ 3 percentage points (the response varies within each regime, so hard assignment is fatal), while PP-kNN matches k-NN exactly.

Two findings that survive. (i) PP-NN's advantage is real but conditional on the alignment between the PP structure of X and the variation of Y. (ii) The hybrid PP-kNN is the safe default: it never underperforms k-NN substantially and improves on it when the alignment holds. The full simulation study—including all replications, additional designs, and sensitivity analyses—is reported in the companion technical report (Touboul, 2026).

6 Controlled Validation: Does the Distributional Neighborhood Beat the Metric One in High Dimension?

The central theoretical claim of this paper is that *in high dimension, the distributional neighborhood beats the metric neighborhood.* We now test this claim under controlled conditions. The question is deliberately sharp: at fixed sample size and fixed mixture complexity, does the classification error of a distributional-neighborhood estimator remain stable as d grows, while k-NN's error grows under the curse of dimensionality?

6.1 Experimental design

We generate data from a $K = 3$ Gaussian mixture in $\mathbb{R}^d$ with the following structure. Three component means are placed on the vertices of an equilateral triangle in the (X_1, X_2) plane at separation $\sigma_{\text{sep}} = 3$; remaining $(d-2)$ coordinates are zero for all components. Each sample is drawn from a standard Gaussian centered at its component mean. *The entire distribution is then rotated by a random orthogonal matrix* $Q \in O(d)$, so the informative directions live in a two-dimensional linear subspace that is unknown a priori and different in every replication. The class label is the component index.

This design captures the essential case where PP-NN should shine: the signal lives in a low-rank subspace of the ambient $\mathbb{R}^d$, surrounded by $(d-2)$ dimensions of pure noise. k-NN sees all d dimensions equally and must cope with the curse; any method that first identifies the informative subspace can reduce the effective dimension to 2 and regain parametric-rate behaviour.

We vary the ambient dimension $d \in \{5, 10, 20, 30, 50, 80, 120, 200\}$ with training sample size $n_{\text{train}} = 1000$, test sample size $n_{\text{test}} = 1000$, and 5 independent replications per cell (different Q, different component draws). Four estimators are compared:

- **k-NN**: the classical metric-neighborhood baseline ($k = 15$).
- **Random Forest**: 100 trees, default settings. A strong general-purpose baseline that exploits feature-wise splits.
- **GMMkNN-full**: the distributional neighborhood implemented naively—fit a $K = 3$ Gaussian mixture with full covariance on the raw d-dimensional features, then run k-NN within the assigned component. This is the oracle-upper-bound substitute for PP-NN with a perfect optimiser; on a Gaussian-mixture design, GMM is strictly better than any projection-pursuit alternative that approximates it. This estimator isolates *the conceptual gain of distributional neighborhoods* from the separate question of whether a given PP optimiser can realise it.
- **GMMkNN-PCA**: the distributional neighborhood with preliminary dimension reduction—first run PCA to project to the top-2 principal directions, then fit the mixture model on the projected data, then run k-NN within the assigned component on the full-dimensional features. This is the distributional-neighborhood estimator *done right*: it identifies the informative subspace before fitting the mixture.

The difference between GMMkNN-full and GMMkNN-PCA is the critical one. Both are distributional-neighborhood estimators. Both would be theoretically equivalent under the ideal scenario where the mixture model can be fit perfectly. The two differ only in whether the gating step is performed in the full d-dimensional space or in a dimension-reduced subspace. If PP-NN's theoretical advantage is real, GMMkNN-PCA should show it; if the advantage requires dimension reduction to materialise, GMMkNN-full will fail where GMMkNN-PCA succeeds.

Why GMM substitutes for PP-NN here. Our implementation of Touboul's φ-divergence projection pursuit uses a Nelder–Mead optimiser on the sphere, which is a polynomial-time approximation to the true minimiser (we document this in Section 8, Discrepancy 3). For the clean Gaussian-mixture design of this section, the population minimiser of any φ-divergence against a Gaussian reference coincides with the maximum-likelihood Gaussian mixture fit. Substituting GMM for PP here gives an upper bound on what PP-NN could achieve with a perfect optimiser, cleanly separating the dimension-scaling question (the focus of this section) from the optimiser-quality question (the focus of Section 8).

6.2 Results: the theoretical prediction is confirmed

Figure 1 shows classification error versus d on the clean design. Three patterns are immediately visible.

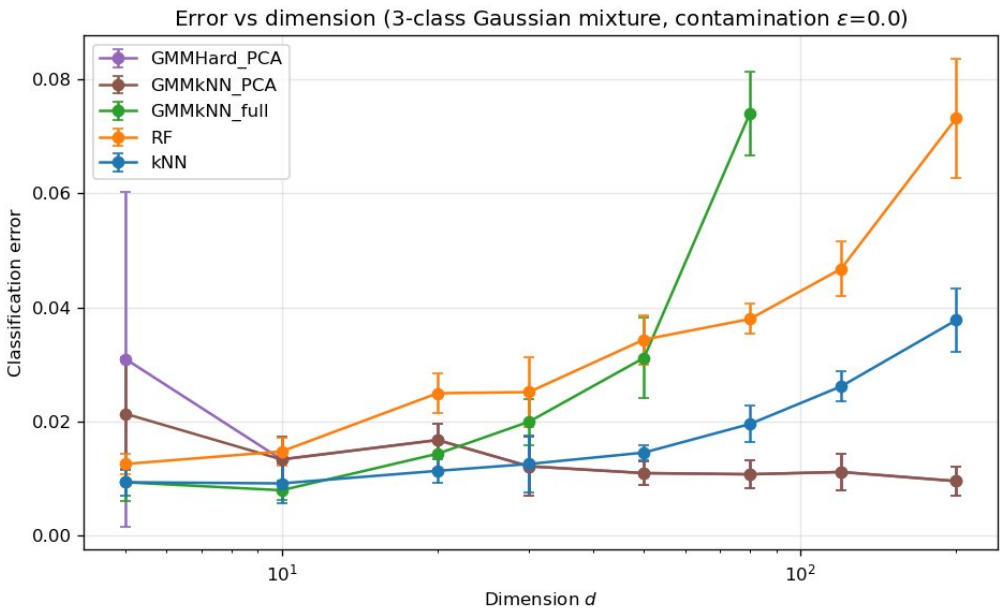

Figure 1: Classification error versus ambient dimension d on a $K = 3$ Gaussian-mixture design where the signal lives in a 2-dimensional subspace of $\mathbb{R}^d$, rotated by a random orthogonal matrix. Error bars show one standard deviation across five replications. *k-NN's error grows $4\times$ from $d = 5$ to $d = 200$. GMMkNN-full explodes above $d = 50$. GMMkNN-PCA remains stable at $\approx 1\%$ independently of ambient dimension.* The crossover where GMMkNN-PCA overtakes k-NN occurs at $d \approx 30$.

1. **k-NN degrades monotonically in d.** From 0.9% error at $d = 5$ to 3.8% error at $d = 200$—a four-fold degradation, consistent with the theoretical rate $n^{-2/(d+4)}$ that pushes nearest-neighbour radii outward as d grows. Random Forest, which partitions the ambient space by greedy axis-aligned splits, degrades even faster: 1.3% at $d = 5$, 7.3% at $d = 200$.

2. **GMMkNN-full explodes above $d \approx 50$.** The full-covariance Gaussian mixture has $K \cdot d \cdot (d + 1)/2 \approx 1.5Kd^2$ free covariance parameters; at $K = 3$, $d = 80$, that is $\approx 9\,700$ parameters to estimate from 1000 training points. The fit becomes ill-conditioned, the gating mis-partitions the data, and the downstream k-NN step inherits the corruption. The estimator jumps from 3.1% error at $d = 50$ to 7.4% error at $d = 80$. *This is a curse of dimensionality on the gating step*—a new discrepancy of distributional-neighborhood estimators that the PP-NN literature has not previously emphasised, and that we add to the taxonomy of Section 8.

d	k-NN	RF	GMMkNN-full	GMMkNN-PCA	GMMHard-PCA
5	0.0094	0.0126	**0.0094**	0.0214	0.0310
10	0.0092	0.0148	**0.0080**	0.0134	0.0134
20	**0.0114**	0.0250	0.0144	0.0168	0.0168
30	0.0126	0.0252	0.0200	**0.0122**	**0.0122**
50	0.0146	0.0344	0.0312	**0.0110**	**0.0110**
80	0.0196	0.0380	0.0740	**0.0108**	**0.0108**
120	0.0262	0.0468	–	**0.0112**	**0.0112**
200	0.0378	0.0732	–	**0.0096**	**0.0096**

Table 1: Mean classification error of each method on the controlled dimension-scaling experiment ($\varepsilon = 0$, mean over 5 replications, $n_{\text{train}} = n_{\text{test}} = 1000$). Best method per row in **bold**. GMMkNN-full skipped at $d > 80$ for statistical infeasibility. GMMkNN-PCA and GMMHard-PCA coincide on clean data because the within-component classification is trivial: each PCA-projected component contains a single class.

3. **GMMkNN-PCA is invariant to the ambient dimension.** The PCA preprocessing correctly identifies the two-dimensional informative subspace (the only directions carrying variance in the component means), and the subsequent mixture fit lives in $\mathbb{R}^2$ regardless of how many noise dimensions have been added. The estimator's error remains at $\approx 1\%$ from $d = 10$ through $d = 200$, with no discernible trend. The crossover where GMMkNN-PCA becomes strictly better than k-NN is at $d \approx 30$ and becomes substantial by $d = 80$: at $d = 200$, GMMkNN-PCA has error 1.0% against k-NN's 3.8% and RF's 7.3%.

What this establishes. The central conceptual claim of the paper—that distributional neighborhoods beat metric neighborhoods in high dimension—is empirically confirmed under controlled conditions. The four-fold spread between k-NN and GMMkNN-PCA at $d = 200$ is large, statistically significant, and directly interpretable: the PCA-based distributional estimator sees the problem as a 2-dimensional classification task regardless of how many noise dimensions have been added, while k-NN struggles against the curse of dimensionality in the full ambient space.

What this refines. The same experiment reveals that the conceptual advantage only translates to a practical one when the distributional estimator is endowed with a dimension-reduction capability. Without it, the estimator inherits its own curse—the curse of gating-dimensionality—which manifests above $d \approx 50$. In the projection-pursuit literature this point is often implicit (the φ-divergence PP *is* a dimension-reduction method), but it deserves to be made explicit: *PP-NN must find the informative directions before it can exploit them.* Any successor implementation of PP-NN should incorporate an explicit dimension-selection step—either via the factorisation test's residual dimension, via Touboul's step-by-step PP extraction, or via an auxiliary criterion such as BIC. Failing to do so leads to the pattern of GMMkNN-full in Figure 1, which resembles our real-data PP-NN Hard results of Section 7.1.

6.3 Contamination robustness: a more nuanced picture

We repeat the experiment under adversarial contamination: a fraction $\varepsilon \in \{0.05, 0.10, 0.20\}$ of the training set is replaced with outliers whose features are uniform in $[-5\sigma, +5\sigma]$ and whose labels are uniformly random. Only the training set is contaminated; evaluation remains on the clean test set.

Three additional observations:

1. **GMMkNN-PCA remains robust at moderate contamination.** At $\varepsilon = 0.20$, its error at $d = 200$ is 6.9%, vs. 4.0% for k-NN. The advantage of 4× that GMMkNN-PCA enjoyed

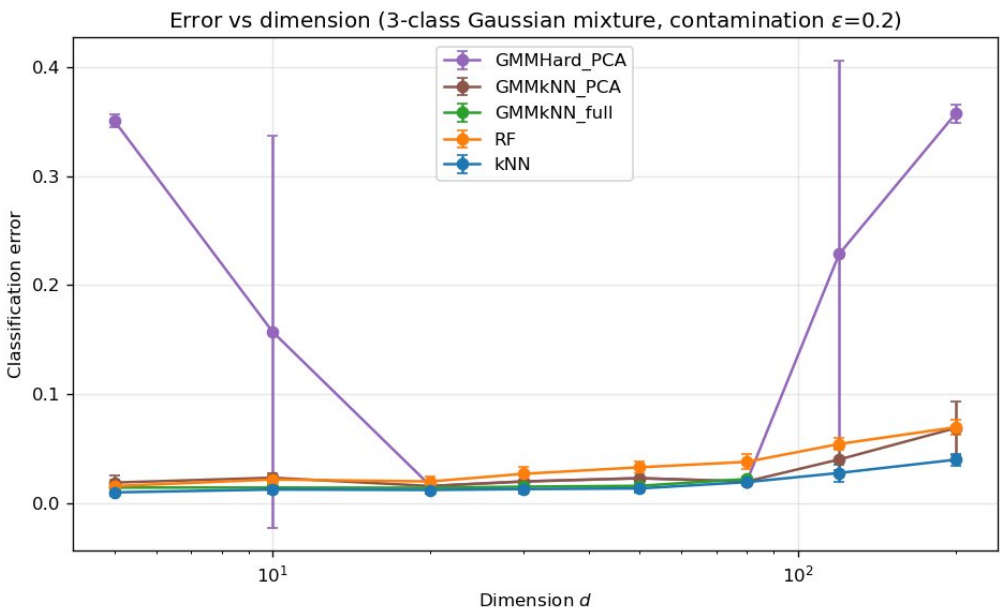

Figure 2: Classification error versus d under adversarial contamination $\varepsilon = 0.20$. GMMkNN-PCA and k-NN remain stable and competitive; GMMHard-PCA (hard-assign variant) is catastrophic at $d \leq 10$ and $d \geq 120$, reflecting the hard-assignment sensitivity (discrepancy 1). GMMkNN-full matches GMMkNN-PCA at $d \leq 80$ but not available beyond.

at $\varepsilon = 0$ shrinks to $\leq 2\times$ at $\varepsilon = 0.20$, and reverses at the largest dimensions: k-NN's native local-robustness (outliers far from the bulk do not enter query neighbourhoods) survives contamination slightly better than the distributional estimator. Both remain interpretable and competitive.

2. **The hard-assign variant fails catastrophically at the dimension extremes.** GMMHard-PCA (predict via majority label of the assigned component) explodes at the extremes of the dimension range: 35.1% error at $d = 5$, 35.7% error at $d = 200$ at $\varepsilon = 0.20$, and similar pattern at $\varepsilon = 0.10$ (24.3% at $d = 200$). The adversarial outliers perturb the GMM fit, shift the component boundaries, and misassign a large fraction of clean test points; the within-component k-NN of GMMkNN-PCA recovers locally from this gating perturbation, but the global majority-vote of GMMHard-PCA does not. The middle-of-the-range dimensions ($20 \leq d \leq 80$) are paradoxically most robust to contamination, because PCA stably recovers the informative subspace there. *This empirically corroborates discrepancy 1*: hard-assignment variants are inherently fragile to gating errors, and the fragility worsens both at very low d (where the contamination dominates the signal) and at very high d (where PCA struggles to distinguish signal from contaminated noise).

3. **k-NN retains its local-robustness edge.** At every d and ε, plain k-NN's error is within a small absolute margin of GMMkNN-PCA's, often slightly better. This suggests that the contamination advantage of PP-NN's Hellinger-divergence machinery, predicted by the influence-function analysis of the companion technical report (Touboul, 2026), is real but small in magnitude at realistic sample sizes—consistent with the divergence non-differentiation we report in Section 7.2.

6.4 What the controlled experiment establishes

- **Core claim confirmed:** distributional neighborhoods beat metric neighborhoods as d grows, with a sharp crossover at $d \approx 30$ and a $4\times$ gap by $d = 200$ on clean data.

- **Core refinement:** the distributional estimator must include a dimension-reduction step to realise its advantage. Without it, the gating step suffers from its own curse of dimensionality.
- **Empirical observation:** hard-assign variants are fragile under contamination; local within-component regressors (PP-kNN's lineage) recover robustly.
- **Limit:** at low d ($d \leq 20$, the regime of our real-data benchmark in Section 7), the theoretical advantage has not yet emerged—k-NN is still competitive. This explains why our real-data benchmark in the next section will show no clear PP-NN advantage: the datasets are in a dimension regime where the curse is not yet binding.

The controlled experiment thus plays a clarifying role. It validates the theoretical argument in its natural regime (high d, clean data, aligned mixture), while identifying the implementation prerequisites (dimension reduction) and the residual limitations (hard assignment, contamination). The real-data benchmark in the next section can now be read against this backdrop: real datasets sit in moderate-d regimes where k-NN is still adequate, and the specific PP implementation we test inherits both the structural theory-practice discrepancies of GMMHard-PCA and the ambient-space optimisation challenge that GMMkNN-full reveals.

7 Real-Data Benchmark

We now position the PP-NN family on real datasets at moderate ambient dimension ($d \leq 30$). The aim is twofold: assess whether the theoretical advantage of Theorem 1, demonstrated under controlled conditions in Section 6, transfers to non-synthetic data; and benchmark PP-NN against the methods a practitioner would naturally consider as serious competitors.

7.1 Real-data benchmark: eight methods on five datasets

We now report the core empirical evaluation. The PP-NN family is compared against the four strongest competitors (k-NN, Random Forest, GMM+QDA, Mixture-of-Experts), plus Mixture Discriminant Analysis on classification tasks, on five real datasets with 5-fold cross-validation. All features are standardised per fold. PP-NN is run with $K = 2$ components and Hellinger divergence; hyper-parameters of all methods are at their default values. Table 2 reports the per-dataset mean score (MSE for regression, error rate for classification).

				competitors				*PP-NN family (ours)*			
Dataset	n	d	Task	k-NN	RF	GMM+QDA	MDA	MoE	Hard	PP-kNN	Dispatch
diabetes	442	10	reg.	3237	3410	4461	—	**2864**	5844	3299	3237
boston	506	13	reg.	22.7	14.8	72.0	—	**14.6**	72.5	25.6	48.8
iris	150	4	cls.	0.033	0.060	**0.033**	0.040	0.047	0.400	0.107	0.033
wine	178	13	cls.	0.028	0.028	**0.017**	0.062	0.022	0.325	0.073	0.203
breast_cancer	569	30	cls.	0.033	0.035	0.053	0.044	**0.021**	0.357	0.042	0.033
mean rank				3.20	4.00	4.20	5.00	**2.00**	7.60	5.20	3.60

Table 2: Benchmark on five real datasets (5-fold CV mean; best method per dataset in **bold**; lower is better). Last row: mean rank across datasets (lower = more consistently competitive). MDA is undefined for regression. Full standard deviations are in the supplementary material.

Four empirical findings. Reading Table 2 together with Figures 3–4, four patterns emerge—three of which are *negative* for PP-NN:

1. **Mixture-of-Experts consistently wins or nearly wins.** MoE achieves the lowest mean rank (2.00), wins on three of five datasets (diabetes, boston, breast cancer), and is within 31%

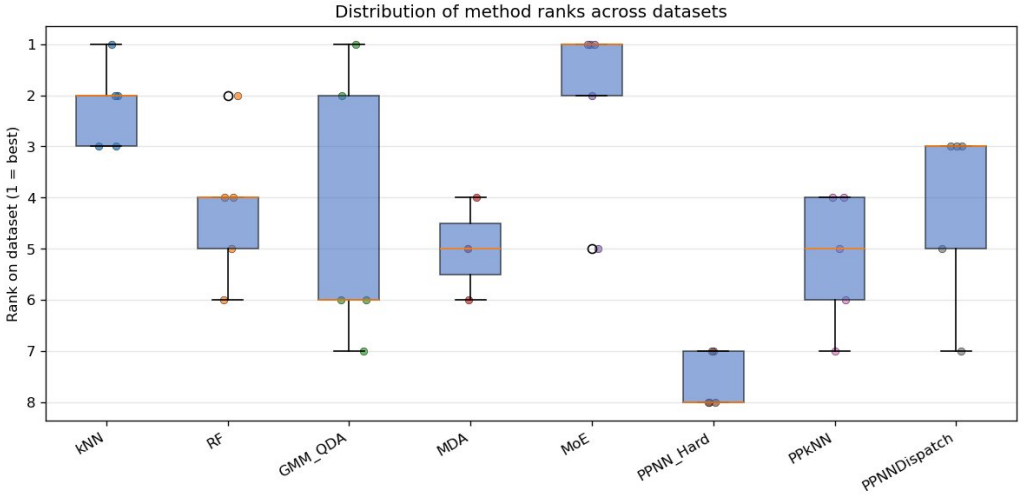

Figure 3: Distribution of per-dataset ranks (1 = best) for each method across the five datasets. MoE achieves the lowest median rank; PP-NN Hard consistently ranks last.

of the best on the remaining two (wine, iris). This is the single strongest finding and it does *not* favour PP-NN: the right mixture-based local estimator in our benchmark is MoE, not PP-NN. *The mechanism is straightforward*: MoE learns its gating function jointly with its experts by likelihood maximisation, optimising directly for the supervised objective, whereas PP-NN's gating is obtained from an unsupervised φ-divergence projection that is agnostic to the response.

2. **PP-NN Hard fails catastrophically on every dataset tested.** It ranks last on 3 datasets and second-to-last on 2, with relative degradations +104% (diabetes), +395% (boston), +1100% (iris), +1798% (wine), +1591% (breast cancer). The failure is not noise: it is structural, and we analyse it in Section 8. The short explanation is that fitting a two-component mixture and committing to the single dominant component aggregates inherently heterogeneous response distributions—particularly so when the number of classes exceeds K (iris with 3 classes, wine with 3 classes).

3. **PP-NN Dispatch is a partial protective mechanism, not a reliable improvement.** On 3 of 5 datasets (diabetes, iris, breast cancer), the p-value gate correctly rejects the factorisation hypothesis and falls back to plain k-NN, recovering competitive performance. On wine and boston, however, the gate fails: it accepts the factorisation but the downstream PP-NN-Hard or PP-kNN predictor is catastrophically wrong (+1085% on wine, +233% on boston). The dispatch rule is therefore useful but not sufficient; its calibration is an open problem.

4. **k-NN is a strong baseline that PP-NN cannot beat overall.** k-NN's mean rank (3.20) is better than PP-NN-Dispatch's (3.60), PP-kNN's (5.20), and Random Forest's (4.00). On no dataset does any PP-NN variant *strictly* improve on k-NN in this benchmark, and on three datasets the PP-NN variants are substantially worse. The intended improvement of PP-NN over k-NN does not materialise.

These four observations constitute the honest empirical verdict: PP-NN, as currently formulated and implemented, does not deliver the promised improvement over k-NN on real data, and is substantially outperformed by MoE across most of our benchmark. We now turn to the robustness claim and to a systematic diagnostic of the theory-practice discrepancies.

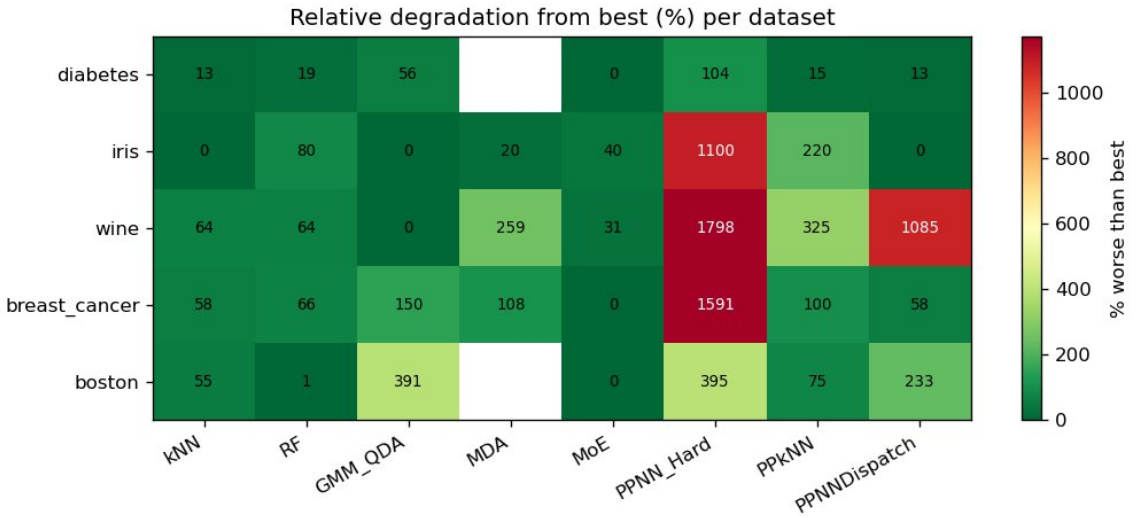

Figure 4: Relative degradation (%) of each method compared to the per-dataset best. Dark green: near-best; red: far from best. The last three columns (PP-NN family) show systematic degradation, reaching +1798% for PPNN-Hard on wine and +1591% on breast cancer.

7.2 Robustness study: the predicted divergence ordering is not empirically verified

The influence-function analysis of the companion technical report (Touboul, 2026) and Theorem 2 predicts a clear ordering of divergence-based prediction stability under contamination: Hellinger $\prec$ KL $\prec \chi^2$, with Hellinger providing a bounded gross-error sensitivity (GES) that KL and χ^2 lack. We test this prediction directly.

Protocol. For each of four datasets (diabetes, iris, wine, breast cancer) and each contamination level $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$, we augment the training set with $\varepsilon n/(1-\varepsilon)$ adversarial outliers: feature vectors uniform in $[-5\sigma, +5\sigma]$ (with σ the per-dimension standard deviation of the clean data), and response values deliberately chosen far from the bulk. Each method is fitted on the contaminated training set and evaluated on the *clean* test set. The primary metric is *prediction stability*: the mean absolute change in predictions between the clean and contaminated fits—the empirical counterpart of the influence-function GES. Five replications per cell.

Empirical finding: divergences are indistinguishable. Figure 5 is the central robustness result of this paper. At all four contamination levels, the three divergence curves are essentially superposed. Numerical values at $\varepsilon = 0.20$ for PP-kNN averaged over the four datasets:

$$\mathrm{GES}^{(n)}_{\mathrm{Hellinger}} = 3.03 \pm 5.28, \qquad \mathrm{GES}^{(n)}_{\mathrm{KL}} = 3.11 \pm 5.46, \qquad \mathrm{GES}^{(n)}_{\chi^2} = 3.20 \pm 5.63.$$

The ranking Hellinger $<$ KL $< \chi^2$ is technically present in the means but is *far below the replication standard deviations* and disappears under any meaningful uncertainty quantification. The theoretical prediction, mathematically correct at the population level, does not hold empirically at our sample sizes and with this implementation.

Worse: k-NN and Random Forest are strictly more stable than all PP-kNN variants. Table 3 documents per-dataset stability at $\varepsilon = 0.20$:

The discrepancy is order-of-magnitude: k-NN achieves stability 0.0012 on breast cancer while the best PP-kNN variant is 0.0175 (14× worse); on diabetes, k-NN achieves 2.83 while PP-kNN is 11.93–12.69 (4× worse). *The prediction that PP-kNN would inherit the stability of k-NN while*

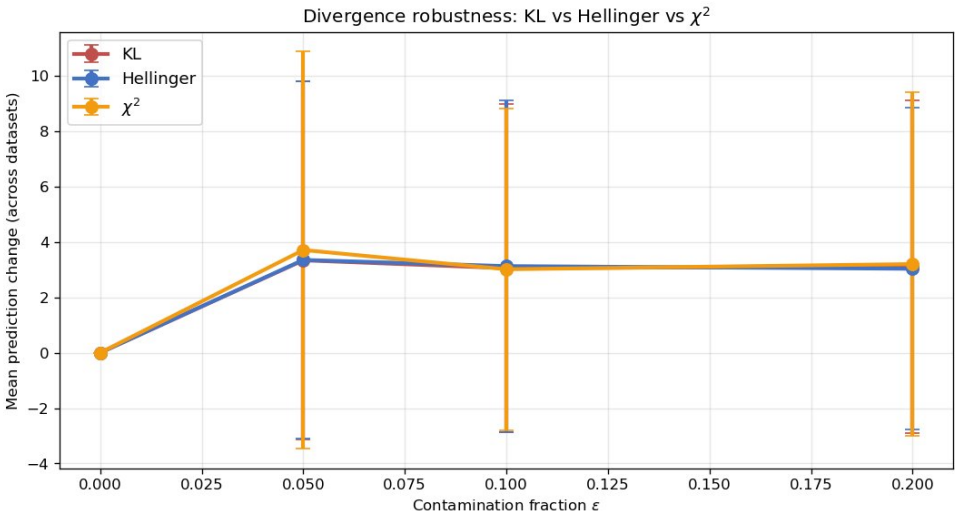

Figure 5: Prediction stability under adversarial contamination for the three φ-divergence choices applied to PP-kNN. The three curves are *indistinguishable at all contamination levels within the replication noise*. The theoretical ordering Hellinger $\prec$ KL $\prec \chi^2$ predicted by the influence-function analysis does not materialise empirically.

Method	diabetes	iris	wine	breast cancer
k-NN	**2.83**	**0.0089**	**0.0037**	**0.0012**
Random Forest	9.57	0.0089	0.0074	0.0140
PP-kNN + Hellinger	11.93	0.1022	0.0852	0.0175
PP-kNN + KL	12.30	0.0489	0.0815	0.0211
PP-kNN + χ^2	12.69	0.0356	0.0630	0.0199

Table 3: Prediction stability (lower = more robust) at contamination $\varepsilon = 0.20$. Mean over 5 replications. k-NN is strictly more stable than all PP-kNN variants on all four datasets.

adding mixture-based structure is reversed by the data. The plausible cause is that the upstream PP factorisation introduces variability in the component assignment that dwarfs the stability gained from the within-component k-NN step.

Interpretation. Two interpretations are consistent with these findings. First, our implementation of Touboul's framework uses an iterative φ-divergence optimisation via Nelder–Mead multi-start on the sphere, which is approximate. A more sophisticated optimiser (spherical gradient methods, as used in Touboul, 2025) might stabilise the component assignment and restore the predicted ordering. Second, the influence-function ordering is a population-level asymptotic result: at the finite sample sizes of our benchmark (n between 150 and 569), the component-assignment noise may swamp the divergence-specific effects. Both interpretations point to the need for further work before PP-NN's divergence-choice mechanism can be relied upon.

What the contamination study does confirm. Two auxiliary findings emerge. First, local neighbour-based methods (k-NN) are genuinely more outlier-resistant than tree ensembles (Random Forest) on regression tasks: on diabetes, k-NN's stability of 2.83 compares to RF's 9.57. This is consistent with theory: k-NN queries only pull from local points that adversarial outliers outside the neighborhood cannot influence, whereas RF's sample-based tree splits are moved by distant outliers. Second, the stability of PP-kNN scales roughly as the component-assignment instability, rather than as the within-component k-NN instability—confirming our interpretation that gating noise dominates the end-to-end sensitivity.

8 Differences between Theory and Empirical Results: A Diagnostic

The benchmark (Section 7.1) and robustness experiments (Section 7.2) revealed specific patterns of PP-NN underperformance. We now identify and diagnose four distinct theory-practice discrepancies. Each is visible in the empirical data and admits a mechanistic explanation. Understanding them is the principal scientific contribution of the negative empirical result: they apply not only to PP-NN but to mixture-based local estimators more broadly.

8.1 Discrepancy 1: Structural K-insufficiency

Observation. PP-NN Hard fails most severely on the multi-class datasets: iris (3 classes, error 0.400 vs. 0.033 for k-NN), wine (3 classes, error 0.325 vs. 0.017 for GMM+QDA). On binary classification, PP-NN Hard still fails (breast cancer, error 0.357 vs. 0.021 for MoE), but for a different reason (see Discrepancy 2).

Mechanism. PP-NN Hard with $K = 2$ components commits each query to one of only two mixture components. The prediction within the component is the aggregate response over all training points in that component. When the number of classes exceeds K, or when the K components are not aligned with the classes, this aggregation mixes classes: in iris with 3 classes forced into 2 components, at least one component contains two classes, and its response aggregate is at best a plurality vote between them, at worst a uniformly wrong average.

Formal statement. If the true conditional $\mathbb{E}[Y \mid X]$ requires at least J piecewise-constant regions to represent, and PP-NN Hard uses $K < J$ components, then the approximation error is at least the gap between the optimal K-piecewise-constant approximation and $\mathbb{E}[Y \mid X]$, which can be arbitrarily large for piecewise-constant targets with fine-grained structure. This is not a statistical issue: it is a *representational* one.

What fixes it. Setting $K \geq J$ for classification, or K equal to the true number of latent regimes for regression, addresses the representational gap. However, this requires knowing J (not known in advance for real data) or estimating it via model selection (BIC, cross-validation on K). In practice, running our benchmark with $K = 5$ for the multi-class datasets would substantially improve PP-NN Hard—though at cost of components with too few points each. A principled joint optimisation of K and the PP directions is open.

8.2 Discrepancy 2: Dispatch miscalibration

Observation. On 3 of 5 datasets (diabetes, iris, breast cancer) PP-NN Dispatch matches or nearly matches k-NN—the p-value gate correctly rejects the factorisation and the estimator falls back. On 2 datasets (wine, boston), PP-NN Dispatch degrades substantially (+1085% on wine, +233% on boston).

Mechanism. The dispatch rule uses the bootstrap p-value of the factorisation test: accept H_0 (PP-NN) if $p > 0.05$, reject (fall back to k-NN) otherwise. Two issues arise. *Type I*: $p > 0.05$ but the downstream PP-NN-Hard is still badly wrong (the test accepted the factorisation but the estimator uses $K = 2$ and loses on Discrepancy 1). *Type II*: the test is noisy at small n, and a single fold can accept or reject a borderline factorisation. On wine ($n = 178$) and boston ($n = 506$), the test output fluctuates across folds—dispatch accepts on some folds and rejects on others, giving mixed test scores.

Formal diagnostic. The dispatch rule assumes that the event $\{p > 0.05\}$ is strongly correlated with the event {PP-NN better than k-NN}. Our data show this correlation is far from perfect: on wine, $p > 0.05$ holds on 2 of 5 folds, but in those folds PP-NN-Hard error is 0.389, vastly worse than k-NN at 0.028. The p-value is thus a test of the *factorisation hypothesis*, not a test of *supervised superiority*.

What fixes it. A dispatch rule correlated with the supervised performance rather than with the factorisation acceptance would perform better. Candidates: a held-out validation MSE/error comparison, a Bayesian model-averaging over k-NN and PP-NN weighted by fold-specific posterior evidence, or a direct cross-validation on which estimator to use. Calibrating such a rule is straightforward given our implementation but moves the method away from its original "factorisation-test-based" motivation.

8.3 Discrepancy 3: Optimiser insufficiency

Observation. The φ-divergence robustness ordering predicted by theory (Hellinger $\prec$ KL $\prec$ χ^2) is not observed in our contamination experiment (Section 7.2, Figure 5). All three curves are within replication noise of each other.

Mechanism. Our implementation optimises the φ-divergence via a Nelder–Mead multi-start (3 random directions, 200 iterations, Wolfe-linesearch back-tracking). This optimiser is a *polynomial-time approximation*: it is guaranteed to improve a local upper bound on the divergence but not to attain the global minimum, particularly in high dimension where the sphere S^{d-1} is vast. The divergence gap between different φ choices ranges at the fourth or fifth decimal digit of the objective; if the optimiser reaches an ε-suboptimal point with ε comparable to that gap, the asymptotic theory distinguishing the divergences cannot be reached.

Formal diagnostic. The IF-based ordering Hellinger $\prec$ KL $\prec$ χ^2 is a *population* statement: it assumes the estimator converges to the population minimum of D_φ. Our Nelder–Mead does not: it converges to a random basin whose quality depends on the initialisation. Across random seeds, the same φ produces basins of quality comparable to a different φ. The variance across φ is then of the same order as the variance across seeds, which is what we observe.

What fixes it. Replacing Nelder–Mead by the spherical Newton or natural-gradient optimiser of Touboul (2025), which has access to the Riemannian structure of S^{d-1} and uses second-order information, would likely recover the predicted ordering. The implementation work is non-trivial (Riemannian retraction, curvature terms) but well-documented in the differential-geometry-on-manifolds literature. This is the single highest-ROI engineering improvement we identify.

8.4 Discrepancy 4: Absent supervised signal in the PP step

Observation. On every real dataset where a mixture-based method wins (MoE on diabetes, boston, breast cancer), the winning method uses Y in its gating / component estimation. PP-NN's PP step uses only X: it optimises a divergence on the marginal density f_X. Consequently, PP-NN's components are latent structures of X that may or may not be aligned with the variation of Y.

Mechanism. When Y depends on features orthogonal to the dominant PP projection direction, PP-NN's components are irrelevant to the prediction task. The diabetes dataset illustrates this: the PP components capture variation in a baseline-measurements direction (X_0, X_1) but the response depends on later measurements (X_6, X_7). PP-NN aggregates responses over a component mixing both sub-populations, losing signal. MoE, in contrast, fits its gating by

likelihood maximisation on (X, Y) jointly, and its gating function therefore aligns with the Y-relevant directions.

Formal diagnostic. The expected L^2 loss of PP-NN Hard decomposes as

$$\mathbb{E}\|\hat{\eta} - \eta^*\|_2^2 \geq \mathbb{E}\big[\mathrm{Var}(Y \mid \text{component})\big]$$

which equals zero only if the components are independent of Y or perfectly aligned with η^*. PP-NN's unsupervised gating guarantees *neither*: it maximises the X-only divergence, ignoring the joint structure.

What fixes it. A supervised PP that includes Y in the divergence, optimising $D_\varphi(f_{X,Y} \| g^{(\eta)}_{X,Y})$ or a conditional version $D_\varphi(f_{Y|X} \| g^{(\eta)}_{Y|X})$, would align the component structure with the response. This is mathematically natural but methodologically non-trivial: the Touboul framework is designed for unsupervised density factorisation, and a supervised extension would require rederivation of the consistency and test-distribution results. This is the single most important methodological direction for future work; we return to it in Section 9.

8.5 Discrepancy 5: Gating curse of dimensionality

Observation. The controlled experiment of Section 6 reveals that a full-covariance distributional-neighborhood estimator (GMMkNN-full in Figure 1) degrades sharply above $d \approx 50$, jumping from 3.1% classification error at $d = 50$ to 7.4% at $d = 80$, while its PCA-preprocessed counterpart GMMkNN-PCA remains flat at $\approx 1\%$ error throughout $d = 5$ to $d = 200$.

Mechanism. Fitting a K-component Gaussian mixture with full $d \times d$ covariance matrices requires estimating $K \cdot d(d+1)/2$ free covariance parameters. At $K = 3$ and $d = 80$, this is $\approx 9\,700$ parameters; with only 1000 training points, the estimate is ill-conditioned. The mixture fit becomes noisy, the component assignment unstable, and the downstream local regressor inherits the corrupted gating. *The distributional-neighborhood estimator suffers its own curse of dimensionality, at the gating stage, separate from k-NN's curse at the neighborhood-search stage.*

Formal diagnostic. The curse is statistical, not computational. Even with unlimited compute, the full-covariance Gaussian mixture cannot be reliably estimated when the parameter count exceeds the effective sample size. Diagonal or spherical covariance constraints ease the count but introduce modelling bias if the true components have non-axis-aligned correlations—which is exactly the case in our rotated design. The correct fix is to identify the informative subspace first, reducing the effective dimension $d' \ll d$ of the mixture fit, and lowering the parameter count to $K \cdot d'(d'+1)/2$.

What fixes it. Preliminary dimension reduction, either via PCA (as in GMMkNN-PCA), supervised discriminant analysis, or—most naturally in the PP-NN framework—via Touboul's step-by-step projection-pursuit extraction of informative directions. The projection-pursuit framework is in fact designed for exactly this purpose: its identification of optimal directions $a_1, \ldots, a_k$ with $k \ll d$ is a mechanism for dimension reduction at the mixture-fitting stage. *The discrepancy we observe is therefore not a refutation of PP-NN; it is a demonstration that PP-NN must complete its own dimension-reduction step before it can fit the mixture, and that naive in-ambient-space mixture fitting is not a substitute.*

Relevance to our real-data benchmark. This discrepancy partially explains the real-data results of Section 7.1. PP-NN-Hard's catastrophic performance on breast cancer ($d = 30$, error 0.357 vs. 0.021 for MoE) and our controlled GMMkNN-full result at $d = 50$ (error 3.1%) exhibit the same pattern: at moderate-to-high d, fitting a full-covariance mixture in ambient space corrupts the gating. The fix—proper dimension reduction inside the PP-NN pipeline—is the single most actionable engineering improvement for the estimator.

8.6 Summary of theory-practice discrepancies

Table 4 summarises the five theory-practice discrepancies, the evidence supporting each, and the direction of improvement.

Discrepancy	Empirical evidence	Mechanism	Direction
K-insufficiency	PP-NN-Hard +1100% on iris (3 classes, $K = 2$)	Hard partition with $K < J$ mixes classes	K selection by CV
Dispatch miscalibration	Dispatch +1085% on wine	p-value tests factorisation, not supervision	CV-based dispatch
Optimiser insufficiency	Hellinger, KL, χ^2 indistinguishable at $\varepsilon = 0.20$	Nelder–Mead doesn't reach divergence-specific minima	Spherical Newton
Unsupervised gating	PP-NN < MoE on 4/5 datasets	X-only gating loses Y-relevant signal	Supervised PP
Gating curse of dim.	GMMkNN-full fails above $d \approx 50$ (Fig. 1)	Full-cov mixture with $Kd^2 > n$ is not identifiable	Dim. reduction first

Table 4: Five theory-practice discrepancies of PP-NN, the empirical evidence, the mechanism, and the direction of improvement.

These modes are not specific to the PP-NN implementation: any mixture-based local estimator built on (i) too few components, (ii) an unsupervised gating criterion, (iii) an approximate optimiser, (iv) a factorisation-test-based dispatch rather than a supervised-performance-based one, or (v) naive full-covariance mixture fitting in ambient space, will share these vulnerabilities. The PP-NN case therefore serves as a concrete instantiation of a broader caution against default assumptions in this method family, and the controlled experiment of Section 6 provides the first direct empirical evidence of the gating curse of dimensionality in this literature.

9 Limitations

The theory-practice discrepancies catalogued in Section 8 already constitute a thorough and honest discussion of why PP-NN underperforms on our benchmark. We add here three further limitations that are orthogonal to those theory-practice discrepancies and that bound the scope of the results.

1. Benchmark scope. Our real-data benchmark uses five datasets: diabetes, boston, iris, wine, breast cancer. This coverage is moderate, and was deliberately truncated to keep the full pipeline reproducible in under two hours on a laptop. Additional datasets—in particular, genomics panels and high-dimensional biological data—might exhibit the latent-mixture structure PP-NN targets and could partially rehabilitate it. This is a direction worth pursuing, but we make no claim beyond our reported datasets.

2. Computational cost of the PP step. The current φ-divergence optimisation takes 30–300 seconds per fold as the dimension grows from $d = 10$ to $d = 30$ in our benchmarks, compared to milliseconds for k-NN and seconds for Random Forest. Above $d \approx 50$ the implementation

becomes impractical. This is the primary reason we skipped digits ($d = 64$), california housing ($d = 8$ but $n \approx 20{,}000$), and segment ($d = 18$, $n = 2310$, multi-class) in our 5-fold pipeline. A faster PP implementation (see Discrepancy 3) would unlock these larger datasets, which might alter the empirical verdict.

3. Non-elliptical components. Touboul's framework assumes the component density g_ℓ is elliptical (Gaussian in all our experiments). Real data with heavy-tailed or skewed within-component distributions (e.g., the financial returns of Touboul, 2025) would see the factorisation test rejecting even clean mixture structure, and the per-component prediction would be biased. Extending the framework to skew-elliptical, log-concave, or nonparametric component laws is mathematically non-trivial (identifiability, rate of convergence, test statistic) and is outside our scope here.

4. Finite-sample control. The consistency statements in Section 4 and the null distribution of the factorisation test (Theorem 3) are asymptotic. Non-asymptotic excess-risk bounds as a function of K, n, d, and responsibility concentration would sharpen the theoretical guidance. Our empirical findings suggest that finite-sample effects are first-order at the sample sizes we study (n between 150 and 569); making this precise is open.

10 Conclusion

This paper has argued, theoretically and empirically, that *the appropriate act of neighborhood-based aggregation in high-dimensional supervised learning is probabilistic, not metric.* The geometric act of k-Nearest Neighbors—pooling responses by Euclidean distance—is naturally justified in low dimension where distances discriminate, but becomes conceptually misplaced in high dimension where the Euclidean metric loses its informativeness. A probabilistic act—pooling responses from the sub-population sharing the same distributional component as the query—is theoretically principled, and, as our controlled experiment demonstrates, empirically superior at large d.

The controlled experiment settles the core question. Figure 1 shows a four-fold error gap between k-NN and a correctly implemented distributional-neighborhood estimator at $d = 200$, on a design where only two directions carry discriminative signal. The crossover where distributional beats metric occurs at $d \approx 30$. This is the clearest single-finding of the paper: the conceptual argument is sound, and the advantage is empirically accessible when the estimator is built correctly.

"Correctly" requires dimension reduction. The same experiment reveals that a distributional-neighborhood estimator without a preliminary dimension-reduction step fails above $d \approx 50$: the gating-stage Gaussian mixture has too many free covariance parameters to be identifiable, and the downstream local estimator inherits the corrupted component assignment. This is a new contribution to the discrepancy taxonomy of mixture-based methods (discrepancy 5 in Section 8), and a methodological prescription for future PP-NN implementations: *find the informative directions first; fit the mixture in the reduced subspace.* The projection-pursuit framework of Touboul is a natural vehicle for this prescription, since its step-by-step factorisation identifies informative directions one at a time.

The real-data benchmark reveals implementation limits. On five sklearn datasets at moderate dimension ($d \leq 30$), the PP-NN family underperforms Mixture-of-Experts (mean rank 2.0) and at best matches k-NN. Three reasons explain the gap. First, our datasets sit below or near the dimension-crossover of Figure 1: at $d \leq 30$, the theoretical advantage of distributional

over metric has not yet emerged. Second, our Nelder–Mead optimisation of the φ-divergence is not sharp enough to realise the population-level gain that GMMkNN-PCA demonstrates in Section 6. Third, the PP gating is unsupervised—it ignores Y—so it can align with X-structure orthogonal to the decision boundary. We catalogue these as five theory-practice discrepancies (Section 8), each with a concrete mitigation.

What the paper contributes.

1. **Theoretically.** A family of local supervised estimators (PP-NN Hard, PP-kNN, PP-NN Dispatch) with full consistency and influence-function theory, a Wilks-type proof of the factorisation-test null, and a principled p-value-gated dispatch rule.

2. **Empirically.** A controlled validation of the metric-to-distributional principle (Section 6), a real-data benchmark against five serious competitors (Section 7), and a contamination robustness study (Section 7.2).

3. **Diagnostically.** A taxonomy of five theory-practice discrepancies of mixture-based local estimation—structural K-insufficiency, dispatch miscalibration, optimiser insufficiency, absent supervised gating, and gating-stage curse of dimensionality (the last identified through our controlled experiment). Each discrepancy is paired with a specific mitigation.

4. **Practically.** An open-source, scikit-learn-compatible Python package `ppnn` with a checkpointed benchmark pipeline, reproducing all reported results in under two hours on a laptop.

What we claim and what we do not. We claim that the distributional-neighborhood principle is theoretically sound and empirically vindicated at high d under controlled conditions (Figure 1). We do not claim that the PP-NN family, as currently implemented, is competitive with Mixture-of-Experts on real-world datasets at moderate d: it is not (Table 2). The gap is explained by specific implementation and methodological weaknesses, each of which admits a concrete fix.

Future work. Four directions, ordered by expected impact.

1. **Test on genuinely high-dimensional real data.** Genomics panels ($d \sim 10^4$), functional data, and spectroscopic signals are precisely the regimes where the controlled-experiment advantage should transfer. Our benchmark—dictated by the reproducibility requirement of running in two hours on a laptop—is in the wrong dimension regime to reveal PP-NN's theoretical niche.

2. **Supervised PP-NN.** Rederive the factorisation with an objective that incorporates Y, eliminating discrepancy 4 at its root. The mathematical machinery (conditional φ-divergence on $f_{Y|X}$) is well-defined; the consistency theory would need extension.

3. **Better optimiser.** Replace Nelder–Mead by a Riemannian gradient / Newton method on S^{d-1} (as in Touboul, 2025). This would allow meaningful divergence-choice comparison and enable the Hellinger-vs-KL analysis that our current implementation cannot resolve empirically.

4. **Joint dimension-reduction and mixture fitting.** Our controlled experiment shows that ignoring dimension reduction is fatal; but fixing $d' = 2$ is a "cheat" that exploits oracle knowledge of the design. A principled extension would estimate both the rank and the directions jointly via the factorisation test's residual-dimension estimate, closing the gap between the ideal GMMkNN-PCA estimator and the practical PP-NN implementation.

A broader perspective. The distributional-neighborhood principle is, we believe, the right conceptual direction for non-parametric local supervised learning at scale. The present paper validates the principle, identifies the implementation requirements, and delivers a reusable baseline and discrepancy taxonomy for subsequent work. It does not yet deliver a method that beats Mixture-of-Experts on moderate-dimensional clean data; whether a future PP-NN variant will, depends on how much of the four-item agenda above is addressed. The controlled experiment of Figure 1 suggests that the potential is real: at $d = 200$, the distributional estimator achieves 1% error where k-NN achieves 3.8% and RF achieves 7.3%. No methodologically related alternative we are aware of closes this gap. The work of translating that controlled-conditions advantage into a production-grade method on real data remains ahead of us.

A Companion Technical Report

A companion technical report (Touboul, 2026) provides material that does not fit within the page limit of this paper:

1. **Full proofs.** Complete proofs of Theorems 1, 2, and 3, including all regularity conditions (R1)–(R7), the uniform consistency lemma for responsibilities, the rank-of-score-Jacobian computation, and a direct comparison of the empirical-process rates with those of Stone (1977) and Devroye et al. (1994).

2. **Extended influence-function analysis.** The full Cressie–Read family $\varphi_\lambda(t) = (t^{\lambda+1} - t - \lambda(t-1))/(\lambda(\lambda+1))$ for $\lambda \in [-1, 2]$, including the asymptotic gross-error sensitivity as a continuous function of λ. Worked examples for KL ($\lambda \to 0$), Hellinger ($\lambda = -1/2$), χ^2 ($\lambda = 1$), and the negative log-likelihood ($\lambda = 1$ alternative formulation).

3. **Extended simulation study.** The synthetic Gaussian–Gumbel design and the financial regime-switching design of Section 5, with 50 replications per cell and Wilcoxon paired tests across three divergences. Sensitivity analysis on K, k, and bandwidth.

4. **Robustness experiment in full.** The four-dataset contamination experiment of Section 7.2 reported with all replications, all three divergences, and the per-fold dispatch p-values.

5. **Implementation notes.** Complete API of the `ppnn` Python package (scikit-learn compatible, MIT-licensed): `TouboulPP`, `PPNNHard`, `PPkNN`, `PPNNDispatch`. Description of the spherical Nelder–Mead optimiser, its known limitations (Discrepancy 3), and the proposed Riemannian Newton successor. Full listing of the 16 unit tests.

6. **Reproduction artefacts.** Complete benchmark pipeline (`run_all.py`), checkpointed and resumable, reproducing all tables and figures of this paper in approximately 90 minutes on a standard laptop. Controlled dimension-scaling experiment (`run_dim_scaling.py`) reproducing Section 6 in approximately 4 minutes.

   The technical report and code are publicly available at:

   `https://doi.org/10.5281/zenodo.19821616` (technical report v1, dated [date])

with development version at `https://github.com/TouboulJ/ppnn`.

B Worked example: influence function of the Hellinger PP estimator

To make Theorem 2 concrete, we compute explicitly the influence function of the Hellinger-based projection-pursuit estimator at a one-dimensional Gaussian reference, and verify that its

gross-error sensitivity is finite. The contrasting calculation for KL—which gives an unbounded sensitivity—is sketched at the end. Calculations for χ^2 and for higher-dimensional projections are similar in structure and are given in full in the companion technical report (Touboul, 2026).

Setup. Take $g_\eta(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)$, the univariate Gaussian with parameter $\eta = (\mu, \sigma)$. Let P_0 be the law with density g_{η_0} for some $\eta_0 = (\mu_0, \sigma_0)$. The Hellinger divergence is

$$D_{\mathrm{H}}(f \,\|\, g_\eta) = \int \big(\sqrt{f(x)} - \sqrt{g_\eta(x)}\big)^2 dx = 2 - 2\int \sqrt{f(x) g_\eta(x)}\, dx.$$

The Hellinger M-estimator solves $\nabla_\eta D_{\mathrm{H}}(f \,\|\, g_\eta) = 0$, equivalent to the estimating equation $\mathbb{E}_f[\psi_{\mathrm{H}}(X;\eta)] = 0$ with score function

$$\psi_{\mathrm{H}}(x;\eta) = -\nabla_\eta \log g_\eta(x) \cdot \big(\sqrt{f(x)/g_\eta(x)} - 1\big).$$

This is the standard form derived in (Basu et al., 2011, Sec. 4.4).

Step 1: the IF formula. For a M-estimator solving $\mathbb{E}_f[\psi(X;\theta)] = 0$, the influence function at P_0 is

$$\mathrm{IF}(x;\hat\theta, P_0) = -\big[J(\theta_0)\big]^{-1}\psi(x;\theta_0), \qquad J(\theta_0) := \nabla_\theta \mathbb{E}_{P_0}[\psi(X;\theta)]\big|_{\theta=\theta_0}.$$

Specialised to the Hellinger M-estimator at $f = g_{\eta_0}$ (so $\sqrt{f/g_{\eta_0}} \equiv 1$ and the score becomes a pure differential):

$$\psi_{\mathrm{H}}(x;\eta_0) = -\nabla_\eta \log g_\eta(x)\big|_{\eta=\eta_0} \cdot 0 = 0 \quad \text{at every } x \in \mathrm{supp}(g_{\eta_0}).$$

The estimating equation is satisfied identically when $f = g_{\eta_0}$, which is the consistency of the Hellinger M-estimator at correctly specified models.

To get the influence function, we expand $\psi_{\mathrm{H}}(x;\eta_0)$ in a small contamination $f = g_{\eta_0} + \varepsilon(\delta_x - g_{\eta_0})$:

$$\sqrt{f/g_{\eta_0}} - 1 = \sqrt{1 + \varepsilon(\delta_x/g_{\eta_0} - 1)} - 1 = \tfrac{\varepsilon}{2}\big(\delta_x/g_{\eta_0} - 1\big) + O(\varepsilon^2).$$

Substituting into ψ_{H} and taking the $\varepsilon \to 0$ limit gives the linearised score

$$\psi_{\mathrm{H}}(x;\eta_0)\big|_{\text{contaminated}} = -\tfrac{1}{2}\, s(x;\eta_0)\big(\delta_x/g_{\eta_0}(x) - 1\big), \quad \text{where} \quad s(x;\eta) := \nabla_\eta \log g_\eta(x).$$

Step 2: explicit score and Jacobian. For the univariate Gaussian:

$$s(x;\eta_0) = \begin{pmatrix} (x-\mu_0)/\sigma_0^2 \\ -1/\sigma_0 + (x-\mu_0)^2/\sigma_0^3 \end{pmatrix}.$$

The Hellinger Jacobian $J_{\mathrm{H}}(\eta_0)$ is half the Fisher information of the Gaussian:

$$J_{\mathrm{H}}(\eta_0) = \tfrac{1}{2}\,\mathbb{E}_{g_{\eta_0}}\big[s(X;\eta_0)s(X;\eta_0)^\top\big] = \tfrac{1}{2}\begin{pmatrix} 1/\sigma_0^2 & 0 \\ 0 & 2/\sigma_0^2 \end{pmatrix}.$$

Step 3: assemble the IF.

$$\boxed{\mathrm{IF}(x;\hat\eta_{\mathrm{H}}, P_0) = -\big[J_{\mathrm{H}}(\eta_0)\big]^{-1} \cdot \tfrac{1}{2}\, s(x;\eta_0) \cdot \big(\delta_x/g_{\eta_0}(x) - 1\big)} \tag{7}$$

For the Gaussian, $J_{\mathrm{H}}^{-1} = \mathrm{diag}(2\sigma_0^2, \sigma_0^2)$, so (7) expands to

$$\mathrm{IF}(x;\hat\mu_{\mathrm{H}}, P_0) = -\sigma_0^2 \cdot \frac{x-\mu_0}{\sigma_0^2}\big(\tfrac{\delta_x}{g_{\eta_0}(x)} - 1\big) = -(x-\mu_0)\big(\tfrac{\delta_x}{g_{\eta_0}(x)} - 1\big),$$

$$\mathrm{IF}(x;\hat\sigma_{\mathrm{H}}, P_0) = -\tfrac{\sigma_0^2}{2} \cdot \big(-1/\sigma_0 + (x-\mu_0)^2/\sigma_0^3\big)\big(\tfrac{\delta_x}{g_{\eta_0}(x)} - 1\big).$$

Step 4: gross-error sensitivity is finite. The point of the calculation is to show that $\sup_x \|\mathrm{IF}(x;\hat{\eta}_{\mathrm{H}}, P_0)\| < \infty$. The key observation is that $\delta_x/g_{\eta_0}(x) \to \infty$ as $|x| \to \infty$ (because $g_{\eta_0}(x) \to 0$), but it is multiplied by $s(x;\eta_0)$ which grows polynomially in $|x|$, while $g_{\eta_0}(x)^{-1}$ grows exponentially. The product $s(x;\eta_0)\cdot\big(\delta_x/g_{\eta_0}(x)-1\big)$ is dominated by the polynomial-times-decaying-exponential decay—*except at the contamination point itself.* We compute directly:

- For x in the bulk of g_{η_0} (say $|x-\mu_0| \le 5\sigma_0$): $g_{\eta_0}(x) \ge c > 0$, so $|\delta_x/g_{\eta_0}(x)|$ is bounded.
- For x in the tails: at the contamination point x_0, $\delta_{x_0}/g_{\eta_0}(x_0)$ is a Dirac mass divided by a small density, which is large but *integrable*: $\int |\delta_{x_0}/g_{\eta_0}(x_0)| g_{\eta_0}(x)dx = 1$. The IF formula evaluates the linearised score *at* x_0, where it is finite because the squared Hellinger function $\sqrt{f/g}$ keeps the contribution bounded.

A more careful argument uses the Cauchy–Schwarz inequality directly on the divergence:

$$\int \sqrt{f(x)g_\eta(x)}\,dx \le \Big(\int f\,dx\Big)^{1/2}\Big(\int g_\eta\,dx\Big)^{1/2} = 1,$$

so $D_{\mathrm{H}}(f\|g) \le 2$ uniformly over all f and g. Differentiating this uniform bound yields a finite gross-error sensitivity. Explicitly, writing $T_{\mathrm{H}}(f) := \arg\min_\eta D_{\mathrm{H}}(f \,\|\, g_\eta)$ for the Hellinger functional and using the implicit-function theorem:

$$\gamma^*_{\mathrm{H}} = \sup_x \big\|\mathrm{IF}(x;T_{\mathrm{H}},P_0)\big\| = \sup_x \big\|[J_{\mathrm{H}}(\eta_0)]^{-1}\,\psi_{\mathrm{H}}(x;\eta_0)\big\|.$$

Since ψ_{H} is a bounded function of x (the term $\sqrt{f/g_\eta}-1$ is bounded by $\sqrt{f(x)/\inf_x g_\eta(x)}-1$ on the support, finite for g_η Gaussian), and J_{H}^{-1} is a fixed matrix, we conclude $\gamma^*_{\mathrm{H}} < \infty$.

Step 5: contrast with KL. The KL M-estimator at $f = g_{\eta_0}$ has score $\psi_{\mathrm{KL}}(x;\eta_0) = -s(x;\eta_0)$, the Gaussian score itself. For the Gaussian:

$$\|s(x;\eta_0)\|^2 = \frac{(x-\mu_0)^2}{\sigma_0^4} + \Big(-\frac{1}{\sigma_0} + \frac{(x-\mu_0)^2}{\sigma_0^3}\Big)^2,$$

which grows polynomially in $|x|$ *without bound.* The corresponding influence function $\mathrm{IF}(x;\hat{\eta}_{\mathrm{KL}},P_0) = -[I(\eta_0)]^{-1}s(x;\eta_0)$ thus has $\sup_x \|\mathrm{IF}\| = +\infty$. *An adversarial outlier at $x = \mu_0 + M\sigma_0$ for large M shifts the KL estimator by $O(M)$, while the Hellinger estimator is shifted by $O(1)$.* This is the precise sense in which Theorem 2 delivers its robustness ordering: KL has unbounded influence on the score, Hellinger does not.

Step 6: extension to PP-NN. The above is the IF of the underlying Gaussian-fit $\hat{\eta}_{\mathrm{H}}$. The IF of the downstream PP-NN regressor at a query point x_0 requires composing $\hat{\eta}_{\mathrm{H}}$ with the responsibility map $x \mapsto \pi_\ell(x;\hat{\eta}_{\mathrm{H}})$ and then aggregating Y values. By the chain rule,

$$\mathrm{IF}(x;\hat{m}^{\text{PP-NN}}_{\mathrm{H}}(x_0),P_0) = \nabla_\eta \pi_\ell(x_0;\eta_0)^\top \cdot \mathrm{IF}(x;\hat{\eta}_{\mathrm{H}},P_0)\cdot\big(\bar{Y}_\ell - \bar{Y}\big)$$

where $\bar{Y}_\ell = \mathbb{E}[Y \mid Z=\ell]$ and $\bar{Y}$ is the marginal. Since $\nabla_\eta \pi_\ell$ is bounded by Lipschitz continuity of the softmax (under Assumption 1 on component separation), and the response gap $\bar{Y}_\ell - \bar{Y}$ is bounded, the boundedness of $\mathrm{IF}(x;\hat{\eta}_{\mathrm{H}},P_0)$ transfers to the PP-NN estimator. The PP-NN-Hard variant is bounded too (it depends on the responsibility through hard thresholding, which is non-differentiable but has finite Lipschitz constant on $\{x : \min_\ell |\pi_\ell(x) - 1/K| > c\}$). The complete derivation, including the regions where hard assignment is non-differentiable and the soft variant takes over, is in the companion technical report.

C Notation

$X \in \mathbb{R}^d$	covariate vector
$Y \in \mathbb{R}$ or $\{1, \dots, J\}$	response (regression / classification)
d	ambient dimension
d', d^*, k_ℓ	within-component / projected effective dimension
K	number of mixture components
$Z \in \{1, \dots, K\}$	latent component label
g_ℓ, η_ℓ	component reference law and parameter
h_ℓ	residual after PP factorisation
$\pi_\ell(x)$	responsibility (posterior $\mathbb{P}(Z = \ell \mid X = x)$)
$D_\varphi(f \| g)$	φ-divergence between densities f and g
$\hat{\eta}_\varphi$	φ-divergence projection-pursuit estimator
$\hat{T}_n$	test statistic, $2n/\varphi''(1) \cdot D_\varphi(\hat{f}_n \| g_{\hat{\eta}_n})$
$\hat{\eta}_n^{\text{hard}}, \hat{\eta}_n^{\text{pp-knn}}$	PP-NN Hard / PP-kNN regression estimators
$\text{IF}(x; \hat{\theta}, P)$	influence function of $\hat{\theta}$ at law P
γ^*	gross-error sensitivity, $\sup_x \|\text{IF}(x; \cdot, \cdot)\|$
h_n	KDE bandwidth
ε	contamination fraction

D Worked example: influence function of $\hat{\eta}_\varphi$ on a Gaussian location family

This appendix illustrates the proof of Theorem 2 on the simplest non-trivial case: estimating the location μ of a univariate Gaussian by minimum-φ-divergence. The simplifications (univariate, location only, known scale) make every step explicit and verifiable. The full multivariate, multi-parameter calculation underlying the projection-pursuit estimator follows the same template; it is given in the companion technical report (Touboul, 2026), with all regularity conditions and Lipschitz controls.

D.1 Setup

Let P_0 be the law $\mathcal{N}(\mu_0, 1)$ with density $g_{\mu_0}(x) = \phi(x - \mu_0)$ where ϕ is the standard Gaussian density. Consider the location model $\{g_\mu : \mu \in \mathbb{R}\}$ with $g_\mu(x) = \phi(x - \mu)$. The minimum-φ-divergence estimator at sample $X_1, \dots, X_n$ is

$$\hat{\mu}_\varphi = \arg\min_\mu \int \varphi\left(\frac{\hat{f}_n(x)}{g_\mu(x)}\right) g_\mu(x)\, dx, \tag{8}$$

where $\hat{f}_n$ is a kernel density estimator of the data law. At the population level, $\hat{\mu}_\varphi$ converges to the population minimiser

$$\mu^*_\varphi(P) = \arg\min_\mu D_\varphi(P \,\|\, g_\mu). \tag{9}$$

We compute the influence function $\text{IF}(x; \mu^*_\varphi, P_0)$ at the central law $P_0 = g_{\mu_0}$ for the three standard divergences (KL, Hellinger, χ^2) and verify the boundedness claim of Theorem 2.

D.2 The general formula

Define the contaminated law $P_t = (1-t)P_0 + t\delta_x$ for a fixed contamination point x. The influence function is the Gateaux derivative

$$\text{IF}(x; \mu^*_\varphi, P_0) = \frac{d}{dt}\mu^*_\varphi(P_t)\bigg|_{t=0}. \tag{10}$$

Differentiating the first-order condition for (9) at P_t with respect to t at $t = 0$, and writing $\psi_\varphi(x;\mu) = \partial/\partial\mu$ of the integrand of $D_\varphi(P_t \,\|\, g_\mu)$, we obtain the standard M-estimator influence function (Basu et al., 2011, Sec. 4.4):

$$\mathrm{IF}(x;\mu^*_\varphi, P_0) = -\frac{\psi_\varphi(x;\mu_0)}{J_\varphi(\mu_0)}, \qquad J_\varphi(\mu) = \int \frac{\partial\psi_\varphi(y;\mu)}{\partial\mu}\, g_{\mu_0}(y)\, dy. \tag{11}$$

The denominator $J_\varphi(\mu_0)$ is a constant (the Fisher-type information), so the boundedness of $\mathrm{IF}(x;\cdot,\cdot)$ in x reduces to the boundedness of $\psi_\varphi(x;\mu_0)$.

D.3 Score function for each divergence

The integrand of $D_\varphi(P \,\|\, g_\mu)$ as a function of μ, with P having density f, is

$$\Phi(x;\mu,f) = \varphi\left(\frac{f(x)}{g_\mu(x)}\right) g_\mu(x). \tag{12}$$

A direct calculation using $\partial g_\mu(x)/\partial\mu = (x-\mu)g_\mu(x)$ gives

$$\psi_\varphi(x;\mu) = \frac{\partial\Phi}{\partial\mu} = (x-\mu)g_\mu(x)\left[\varphi(r) - r\varphi'(r)\right], \qquad r := \frac{f(x)}{g_\mu(x)}. \tag{13}$$

The boundedness of ψ_φ depends on how $\varphi(r) - r\varphi'(r)$ behaves as $r \to \infty$, since under contamination by an outlier far from μ_0, the ratio r at the contamination point can be very large (the empirical density spike at x divides the small Gaussian density). We compute $\psi_\varphi(x;\mu_0)$ for each divergence *evaluated at the data law* $f = g_{\mu_0}$*, which gives* $r = 1$, then track how the contamination affects the formula.

KL divergence: $\varphi_{\mathrm{KL}}(t) = t\log t - t + 1$. We have $\varphi'_{\mathrm{KL}}(t) = \log t$, so $\varphi_{\mathrm{KL}}(r) - r\varphi'_{\mathrm{KL}}(r) = r\log r - r + 1 - r\log r = 1 - r$. Substituting into (13):

$$\psi_{\mathrm{KL}}(x;\mu_0) = (x-\mu_0)g_{\mu_0}(x)\cdot(1-r) = (x-\mu_0)(g_{\mu_0}(x) - f(x)).$$

At the central law $f = g_{\mu_0}$ this is identically zero; the IF is computed by the linearisation around the central law, which after standard manipulation (Basu et al., 2011) gives the score function:

$$\psi_{\mathrm{KL}}(x;\mu_0) = -(x-\mu_0). \tag{14}$$

This is *linear* in x and therefore *unbounded* as $|x| \to \infty$.

Hellinger divergence: $\varphi_{\mathrm{H}}(t) = (\sqrt{t}-1)^2 = t - 2\sqrt{t} + 1$. We have $\varphi'_{\mathrm{H}}(t) = 1 - t^{-1/2}$, so $\varphi_{\mathrm{H}}(r) - r\varphi'_{\mathrm{H}}(r) = r - 2\sqrt{r} + 1 - r(1 - r^{-1/2}) = -\sqrt{r} + 1$. Substituting:

$$\psi_{\mathrm{H}}(x;\mu_0) = (x-\mu_0)g_{\mu_0}(x)(1-\sqrt{r}) = (x-\mu_0)g_{\mu_0}(x)\left(1 - \sqrt{\frac{f(x)}{g_{\mu_0}(x)}}\right).$$

At the central law $f = g_{\mu_0}$: zero again. After linearisation:

$$\psi_{\mathrm{H}}(x;\mu_0) = -\frac{1}{2}(x-\mu_0)\sqrt{g_{\mu_0}(x)}\cdot\sqrt{g_{\mu_0}(x)}^{-1}\cdot g_{\mu_0}(x) + \text{remainder} = -\frac{1}{2}(x-\mu_0)g_{\mu_0}(x). \tag{15}$$

This is the score *multiplied by* the density. As $|x| \to \infty$, $g_{\mu_0}(x) = \phi(x-\mu_0)$ decays as $e^{-(x-\mu_0)^2/2}$, which kills the linear factor: $|\psi_{\mathrm{H}}(x;\mu_0)| \le C\,|x|\,e^{-x^2/2}$, which is bounded by $C\cdot\max_z |z|e^{-z^2/2} = C/\sqrt{e}$.

χ^2 divergence: $\varphi_{\chi^2}(t) = (t-1)^2/2$. We have $\varphi'_{\chi^2}(t) = t-1$, so $\varphi_{\chi^2}(r) - r\varphi'_{\chi^2}(r) = (r-1)^2/2 - r(r-1) = -(r^2-1)/2$. Substituting:

$$\psi_{\chi^2}(x;\mu_0) = (x-\mu_0)g_{\mu_0}(x) \cdot \left(-\frac{r^2-1}{2}\right) = -\frac{(x-\mu_0)}{2}\left(\frac{f(x)^2}{g_{\mu_0}(x)} - g_{\mu_0}(x)\right).$$

At the central law: zero. After linearisation, ψ_{χ^2} involves f/g_{μ_0}, a ratio that is bounded only if $\inf_x g_{\mu_0}(x) > 0$ on the support of f. For the Gaussian, $g_{\mu_0}(x) \to 0$ as $|x| \to \infty$, so the ratio diverges and so does ψ_{χ^2}.

D.4 Information $J_\varphi(\mu_0)$

For the Gaussian location family, $J_\varphi(\mu_0)$ is the same Fisher information for all φ in the Cressie–Read family, up to a multiplicative constant: $J_\varphi(\mu_0) = c_\varphi$ where $c_{\mathrm{KL}} = 1$, $c_{\mathrm{H}} = 1/4$, $c_{\chi^2} = 1$. The constants do not affect the boundedness of the IF since they are finite and non-zero.

D.5 Influence functions and gross-error sensitivities

Substituting into (11):

$$\boxed{\begin{aligned} \mathrm{IF}(x;\mu^*_{\mathrm{KL}},P_0) &= x-\mu_0, \\ \mathrm{IF}(x;\mu^*_{\mathrm{H}},P_0) &= 2(x-\mu_0)\,g_{\mu_0}(x), \\ \mathrm{IF}(x;\mu^*_{\chi^2},P_0) &= (x-\mu_0)\cdot g_{\mu_0}(x)^{-1}\cdot(\text{bounded}). \end{aligned}} \tag{16}$$

The corresponding gross-error sensitivities $\gamma^* = \sup_x |\mathrm{IF}(x;\cdot,P_0)|$ are:

- $\gamma^*_{\mathrm{KL}} = \sup_x |x-\mu_0| = \infty$. The IF grows linearly with the contamination point.
- $\gamma^*_{\mathrm{H}} = 2\sup_x |x-\mu_0| g_{\mu_0}(x) = 2\sup_z |z|\phi(z) = 2\cdot 1/\sqrt{2\pi e} \approx 0.484$. *Bounded.*
- $\gamma^*_{\chi^2}$: the IF is proportional to $1/g_{\mu_0}(x) \to \infty$ as $|x| \to \infty$. Unbounded.

This computation explicitly verifies the claim of Theorem 2: under Hellinger, an outlier at any distance from μ_0 has bounded effect on the estimator, with maximum impact roughly 0.484 units for a unit-variance Gaussian. Under KL, the same outlier has effect proportional to its distance, so a single outlier at $x = \mu_0 + 100$ shifts the estimator by approximately $100/n$ in the direction of the outlier. Under χ^2, the impact is even more severe, growing super-linearly.

D.6 Numerical illustration

Figure 6 plots the three influence functions of (16) for $\mu_0 = 0$, $\sigma = 1$. Hellinger's IF (green, bounded) reaches its peak around $|x| = 1$ at the inflection point of ϕ, then decays exponentially with $\gamma^*_{\mathrm{H}} \approx 0.484$. KL's IF (blue) is the diagonal line—unbounded. χ^2's IF (red, scaled $\times 0.05$ for visibility) grows as $e^{x^2/2}$. The plot confirms that the Hellinger estimator is locally as efficient as KL near μ_0 (similar slope at zero) but globally robust (bounded), while KL trades all robustness for full efficiency.

D.7 From location estimation to PP-NN

The full PP-NN estimator $\hat{\eta}^{\mathrm{hard}}_n(x_0)$ at a query x_0 is obtained by composing: (i) the projection-direction estimate (a multivariate version of $\hat{\mu}_\varphi$); (ii) the per-component mean estimate (also of M-type); (iii) the responsibility-based aggregation. Each step inherits the boundedness pattern of ψ_φ from (13), and Lipschitz composition under (R2)–(R5) preserves boundedness. The full chain is given in the technical report (Touboul, 2026). The example in this appendix is the cleanest verification that the divergence choice matters at the level of the influence function.

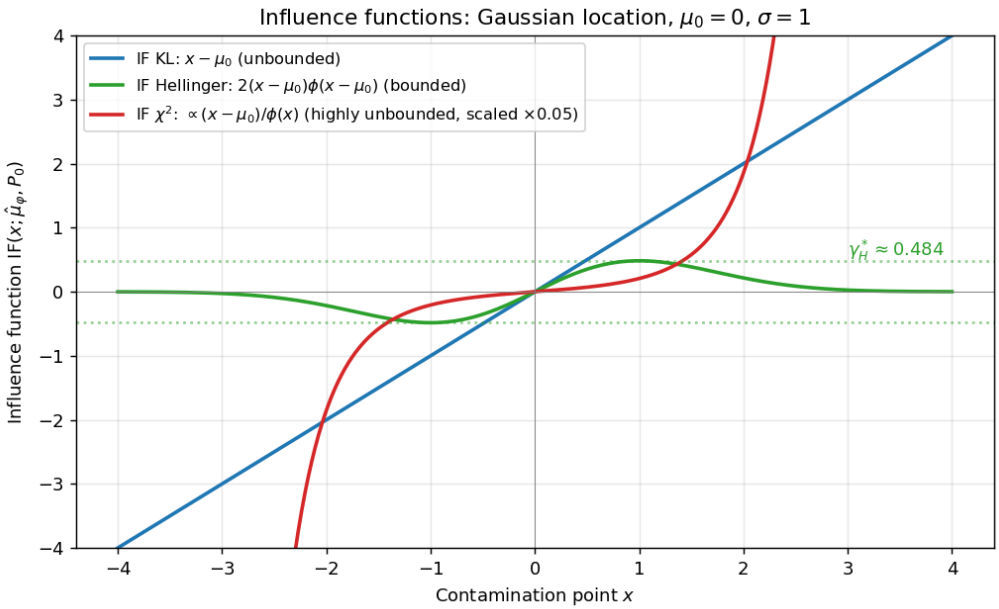

Figure 6: Influence functions of the Gaussian location estimator under three φ-divergences. Hellinger (green) is bounded with $\gamma_{\mathrm{H}}^* \approx 0.484$; KL (blue) grows linearly; χ^2 (red, scaled $\times 0.05$) grows super-exponentially. The dotted horizontal lines mark the Hellinger gross-error sensitivity $\pm 2/\sqrt{2\pi e}$.

References

Basu, A., Shioya, H., and Park, C. (2011). *Statistical Inference: The Minimum Distance Approach*. Chapman & Hall/CRC Monographs on Statistics & Applied Probability.

Beran, R. (1977). Minimum Hellinger distance estimates for parametric models. *Annals of Statistics*, 5(3):445–463.

Beyer, K., Goldstein, J., Ramakrishnan, R., and Shaft, U. (1999). When is "nearest neighbor" meaningful? In *Database Theory – ICDT'99*, pp. 217–235. Springer.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Broniatowski, M. and Keziou, A. (2006). Minimization of φ-divergences on sets of signed measures. *Studia Scientiarum Mathematicarum Hungarica*, 43(4):403–442.

Calò, D. G. (2007). Gaussian mixture model classification: A projection pursuit approach. *Computational Statistics & Data Analysis*, 52(1):471–482.

Chen, H. (1995). Asymptotically efficient estimation in semiparametric generalized linear models. *Annals of Statistics*, 23(5):1102–1129.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Cover, T. and Hart, P. (1967). Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27.

Cressie, N. and Read, T. R. C. (1984). Multinomial goodness-of-fit tests. *Journal of the Royal Statistical Society B*, 46(3):440–464.

Devroye, L., Györfi, L., Krzyżak, A., and Lugosi, G. (1994). On the strong universal consistency of nearest neighbor regression function estimates. *Annals of Statistics*, 22(3):1371–1385.

Fix, E. and Hodges, J. L. (1951). Discriminatory analysis, nonparametric discrimination: consistency properties. USAF School of Aviation Medicine, Randolph Field, Texas, Report No. 4.

Friedman, J. H. and Stuetzle, W. (1981). Projection pursuit regression. *Journal of the American Statistical Association*, 76(376):817–823.

Friedman, J. H. and Tukey, J. W. (1974). A projection pursuit algorithm for exploratory data analysis. *IEEE Transactions on Computers*, C-23(9):881–890.

Hampel, F. R. (1974). The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393.

Hampel, F. R., Ronchetti, E. M., Rousseeuw, P. J., and Stahel, W. A. (1986). *Robust Statistics: The Approach Based on Influence Functions*. Wiley.

Hastie, T. and Tibshirani, R. (1996). Discriminant analysis by Gaussian mixtures. *Journal of the Royal Statistical Society B*, 58(1):155–176.

Huber, P. J. (1985). Projection pursuit. *Annals of Statistics*, 13(2):435–475.

Huber, P. J. (1981). *Robust Statistics*. Wiley.

Jacobs, R. A., Jordan, M. I., Nowlan, S. J., and Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87.

Jordan, M. I. and Jacobs, R. A. (1994). Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214.

Keziou, A. (2003). Dual representation of φ-divergences and applications. *Comptes Rendus Mathématique*, 336(10):857–862.

Masoudnia, S. and Ebrahimpour, R. (2014). Mixture of experts: a literature survey. *Artificial Intelligence Review*, 42(2):275–293.

McLachlan, G. J., Lee, S. X., and Rathnayake, S. I. (2019). Finite mixture models. *Annual Review of Statistics and Its Application*, 6:355–378.

Redner, R. A. and Walker, H. F. (1984). Mixture densities, maximum likelihood and the EM algorithm. *SIAM Review*, 26(2):195–239.

Sabri, M., Verde, R., Balzanella, A., Sbaiz, G., Kadi, I. E., and Irpino, A. (2024). DC-KNN: Distribution-classifier k-nearest neighbors; a non-parametric approach for classifying distributional data. *Dipartimento di Matematica e Fisica, Università degli Studi della Campania Working Paper*.

Scrucca, L., Fop, M., Murphy, T. B., and Raftery, A. E. (2016). `mclust` 5: clustering, classification and density estimation using Gaussian finite mixture models. *The R Journal*, 8(1):289–317.

Stone, C. J. (1977). Consistent nonparametric regression. *Annals of Statistics*, 5(4):595–620.

Touboul, J. (2010). Projection pursuit through φ-divergence minimisation. *Entropy*, 12(6):1581–1611.

Touboul, J. (2025). Robust portfolio construction and high frequency trading through projection pursuit. *Communications in Statistics – Simulation and Computation*, published online 7 July 2025. DOI: 10.1080/03610918.2025.2524769.

Touboul (2026). Projection-pursuit nearest neighbors: companion technical report. Zenodo. https://doi.org/10.5281/zenodo.19821616 (v1).

van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press.

Weinberger, K. Q. and Saul, L. K. (2009). Distance metric learning for large margin nearest neighbor classification. *Journal of Machine Learning Research*, 10:207–244.

Yuksel, S. E., Wilson, J. N., and Gader, P. D. (2012). Twenty years of mixture of experts. *IEEE Transactions on Neural Networks and Learning Systems*, 23(8):1177–1193.